



**Waiting for everyone  
to join**

**TU257  
Information Systems  
Marisa Llorens**

# ER Diagrams 2

Lesson 3 Information Systems

02/10/24

TU257

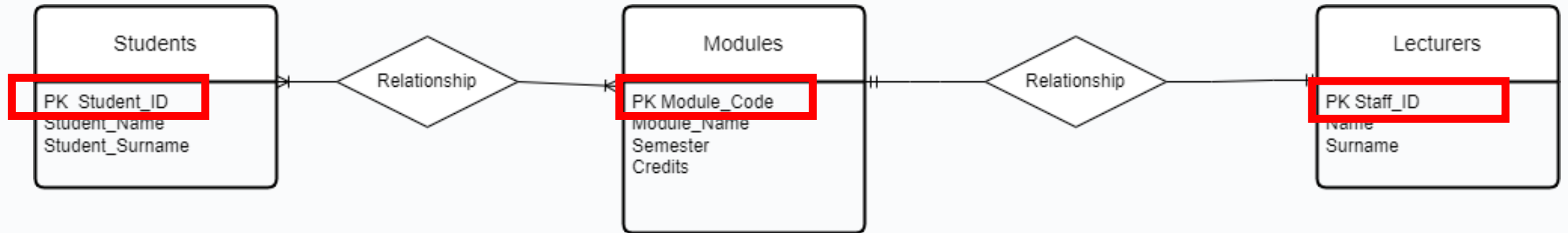
# Lesson Outline

- Foreign Keys
- Many to many relationships and Bridge Entities
- Identifying relationships
- Degree of a relationship
- Key Based ER Diagram

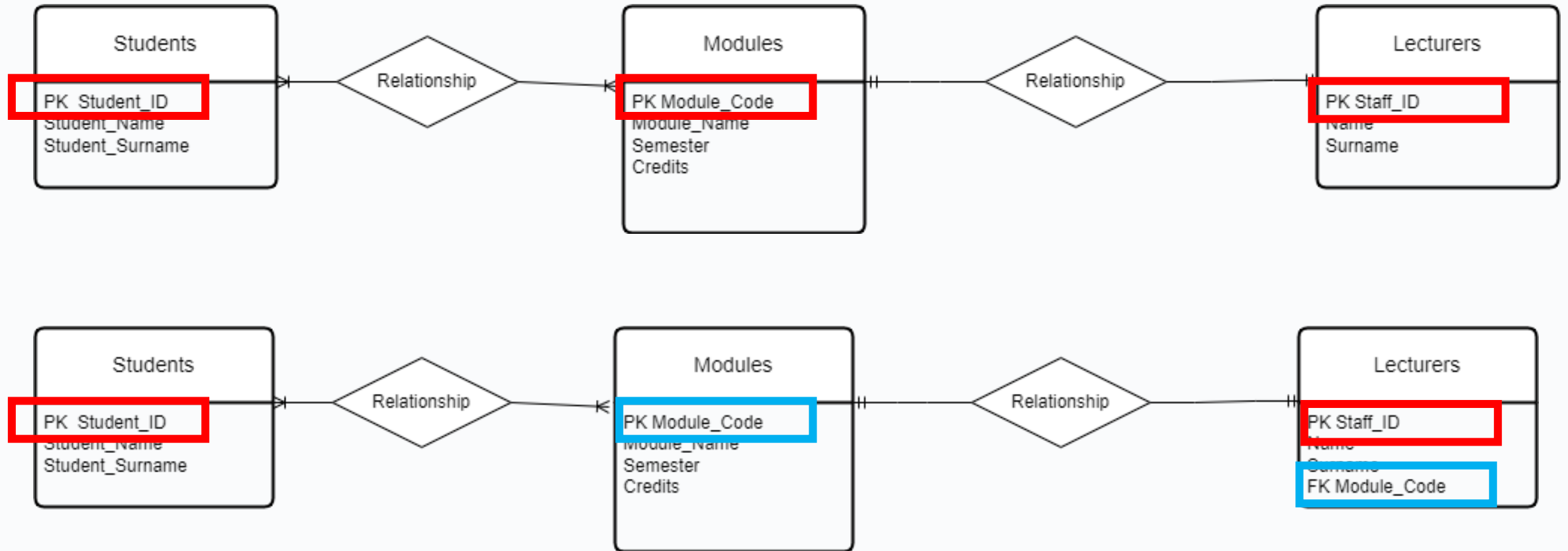
# Foreign Key

- A foreign key is a reference to a key in another table.
- A foreign key is a set of attributes of one table, say **R2**, whose values are required to match values of some candidate key of another table, say **R1**.

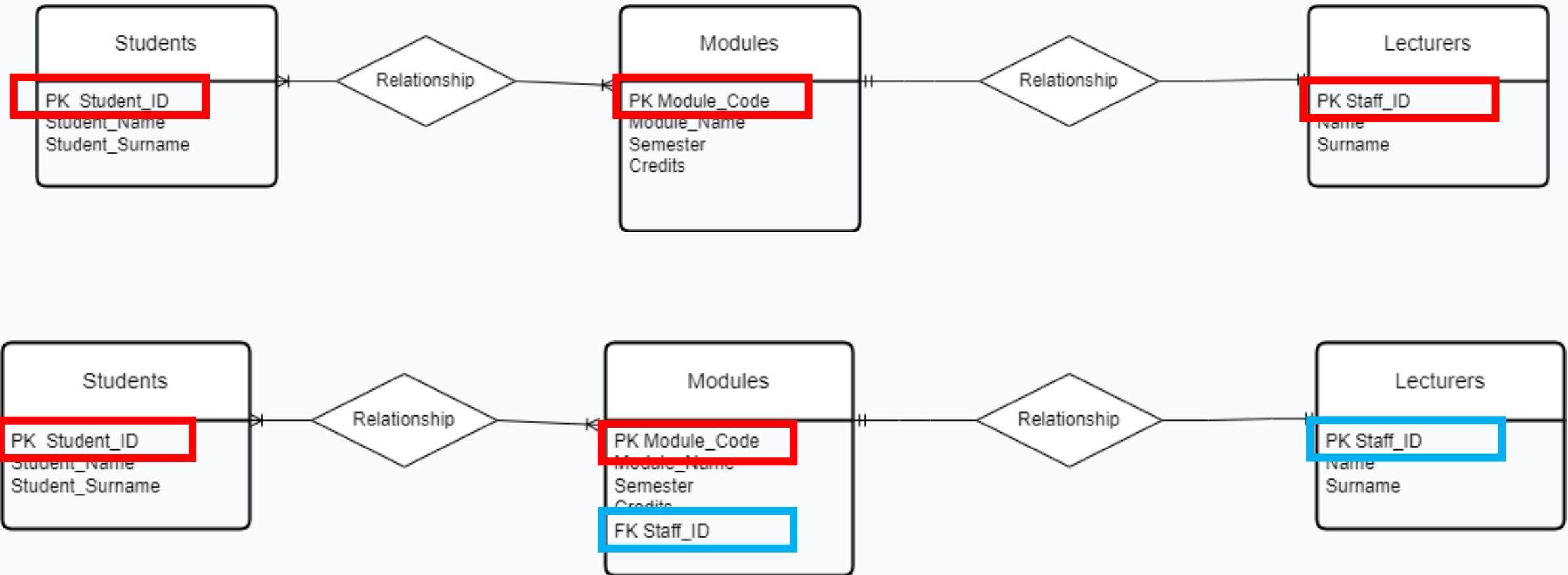
# Example PKs



# Foreign Key One to One v1

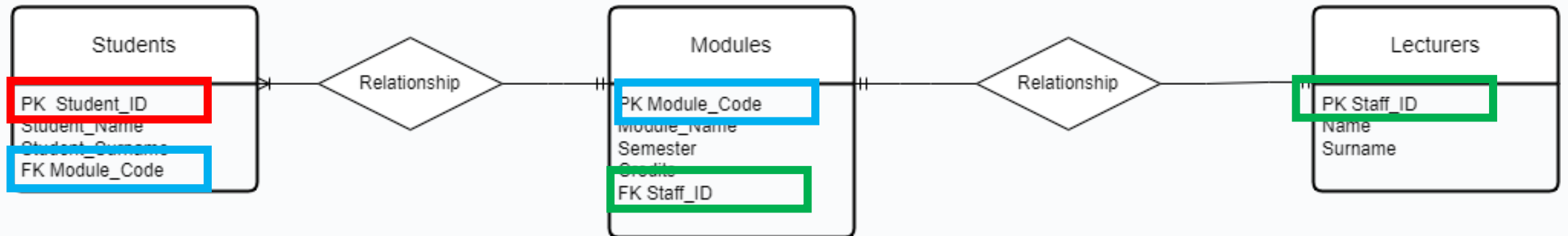


# Foreign Key One to One v2



# Modified Example

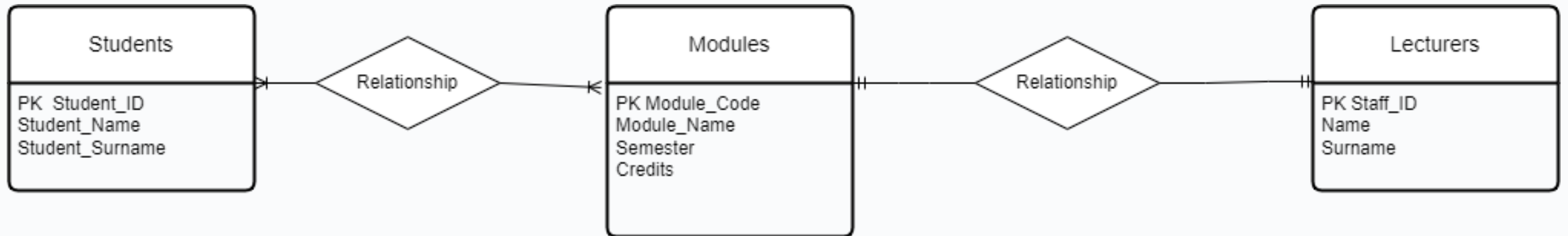
## Foreign Key One to Many



- Case where students can **only** register for one module.
- PK from the one side becomes FK on the many side.



# Foreign Key Many to Many

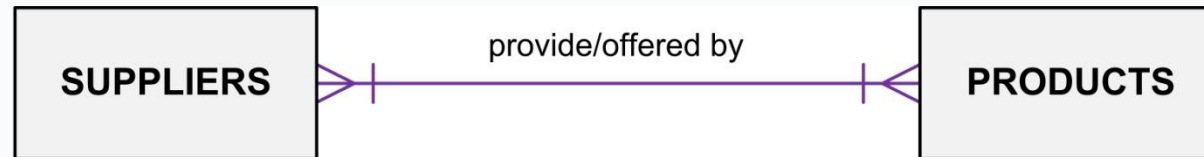


- Case where students can **register for more than one module.**

- When modelling many to many relationships, no FKs can be used directly in the original entities.
- A new entity has to be created.

# Many to Many Relationships Associative (Bridge) Entities

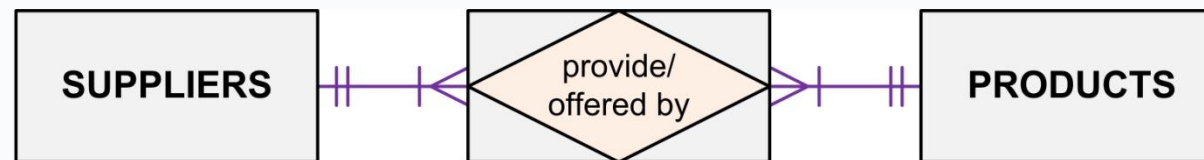
- When an instance of an entity may be related to multiple instances of another entity and vice versa, that is called a '*many-to-many*' relationship.



- While this relationship model is perfectly valid, it cannot be translated directly into a relational database design.

# Many to Many Relationships Associative (Bridge or Weak) Entities

- While this relationship model is perfectly valid, it cannot be translated directly into a relational database design.
- In order to build a relational database that captures this relationship, it is necessary to build a bridge between the two entities that uniquely expresses each relationship instance.

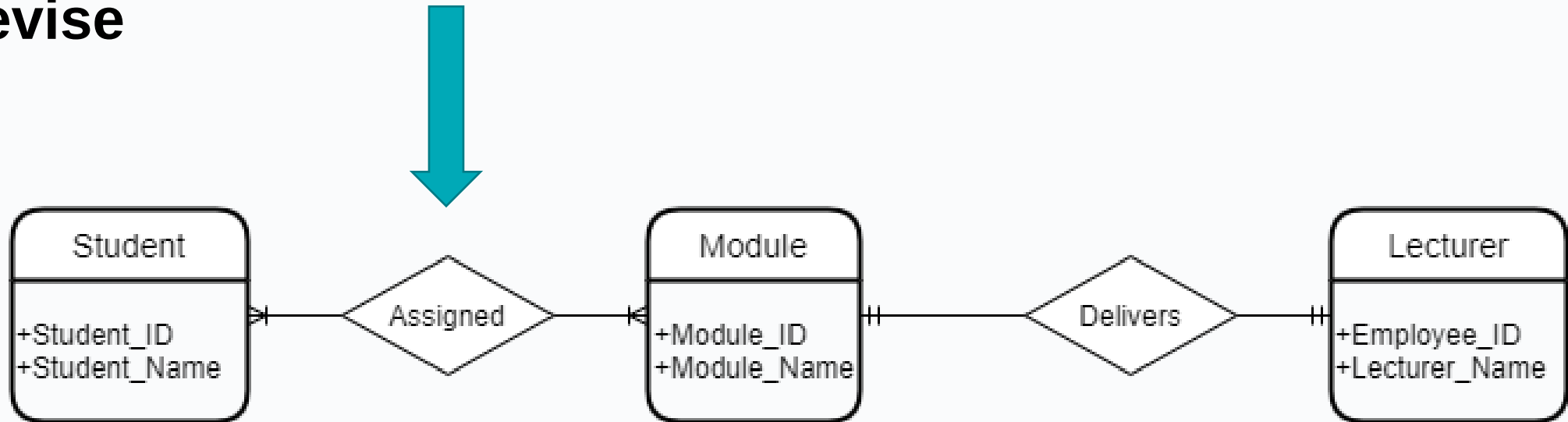


# Example

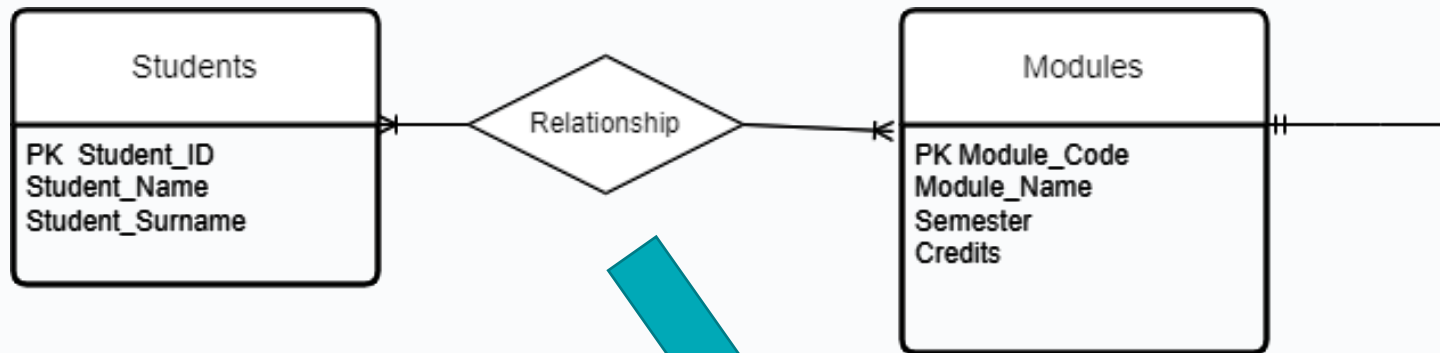
## Student-Module-Lecturer

### Week 2

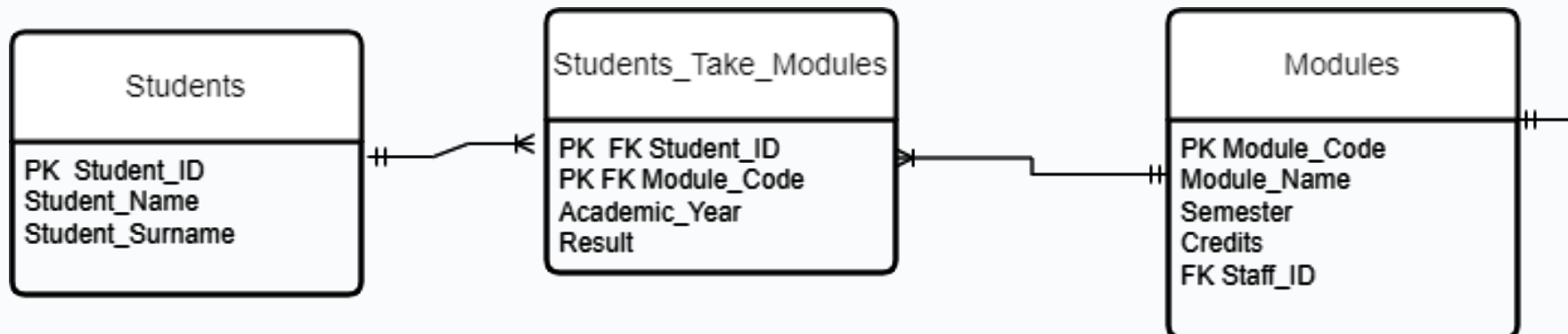
- **Revise**



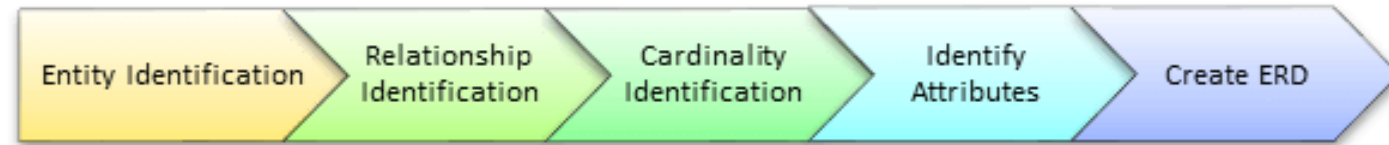
# Associative (Bridge) Entities



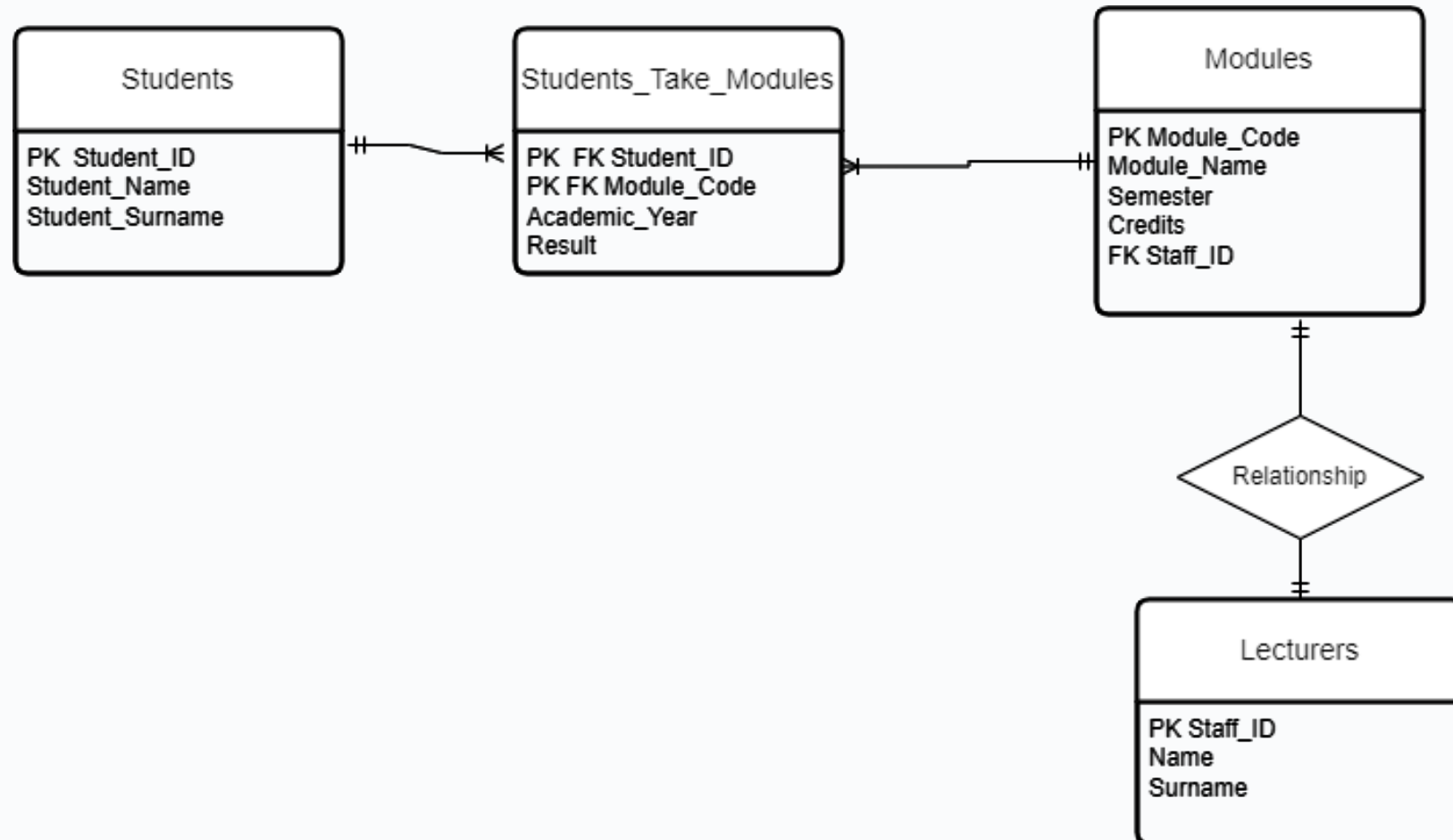
- Many-to-many relationships can cause lots of trouble.
- Get rid of them by defining an intermediate third entity.



# Example

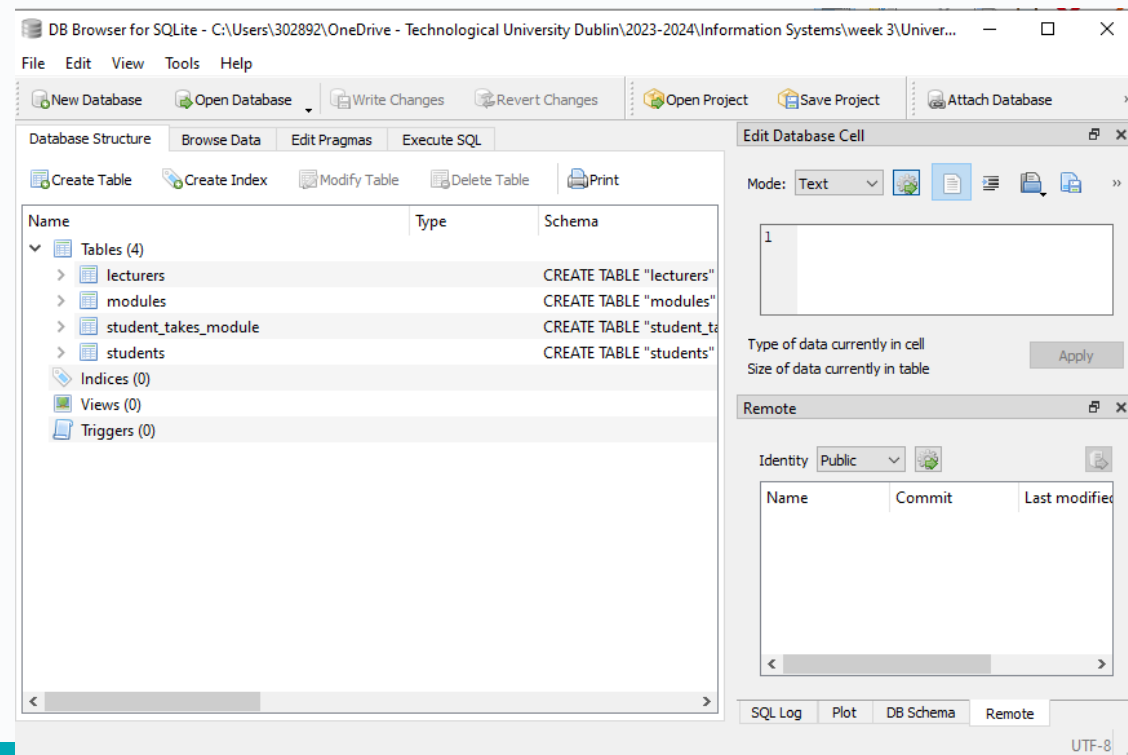


- **Revise**



# Practice

- Open the db provided University.db



# Practice

- Browse the data in the different tables.
- Identify PKs and FKs.



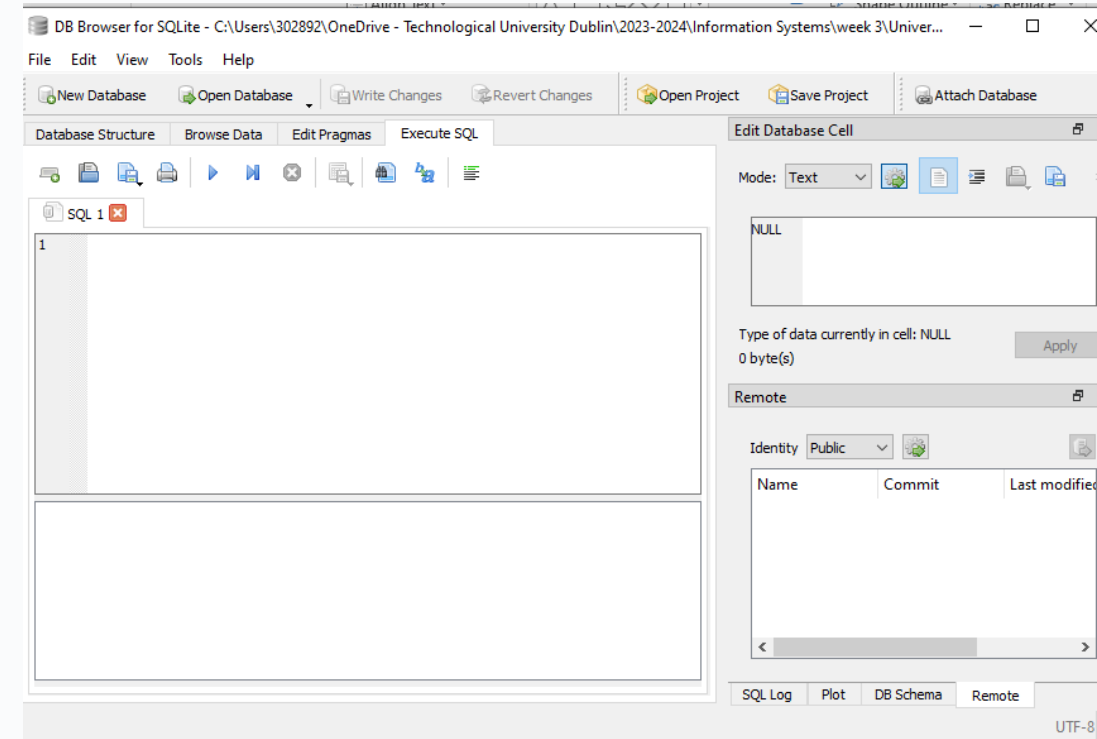
# Practice

- Perform the following queries in the “Execute SQL” Tab

SELECT \* FROM students

SELECT LastName FROM Lecturers

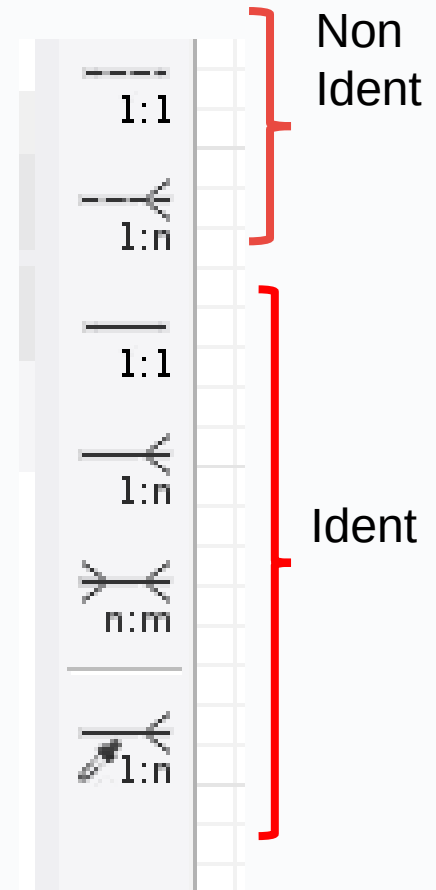
SELECT \* FROM Modules



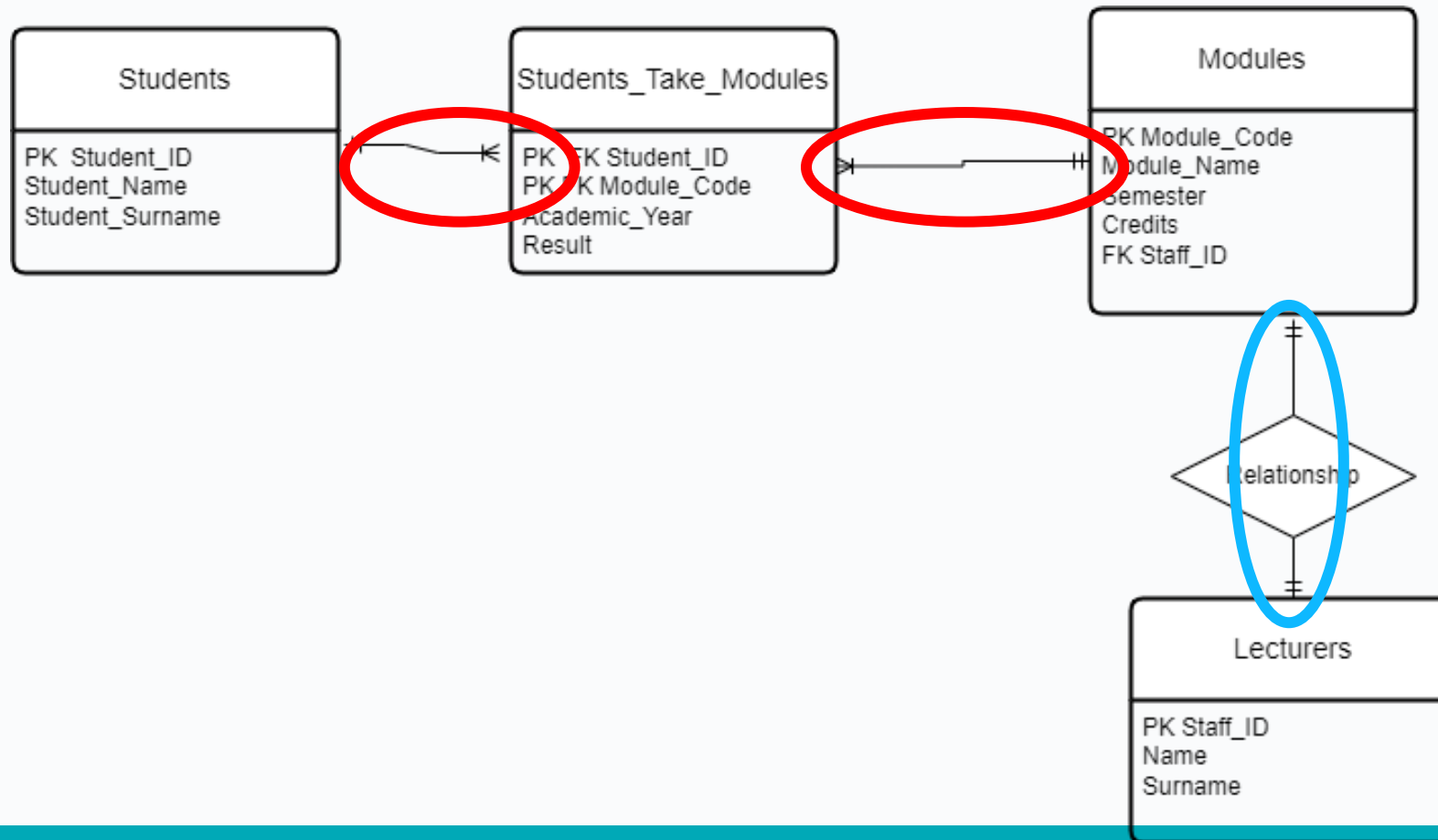
# Identifying Relationships

# Identifying relationships

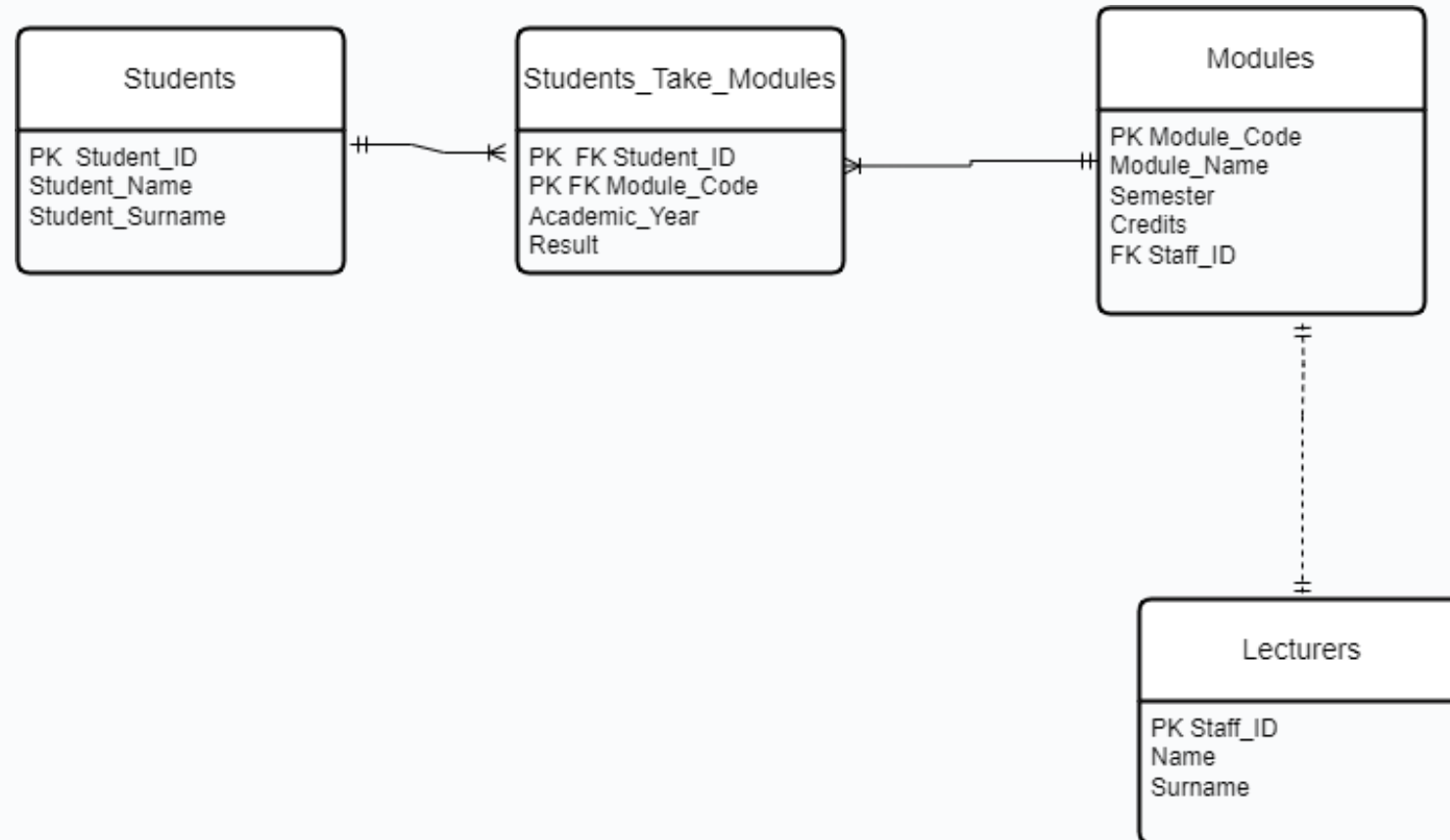
- An identifying relationship means that the child table cannot be uniquely identified without the parent.
- For example, you have this situation in the intersection table used to resolve a many-to-many relationship where the intersecting table's Primary Key is a composite of the left and right (parents) table's Primary Keys.



# Identifying and non identifying relationships



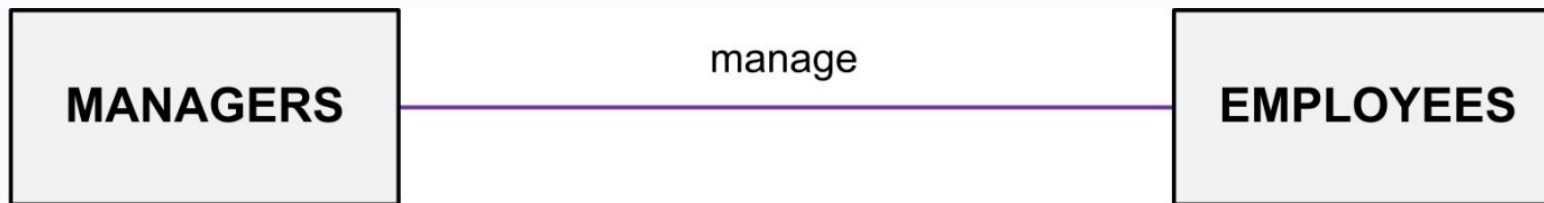
# Identifying and non identifying relationships



# Degree of a relationship

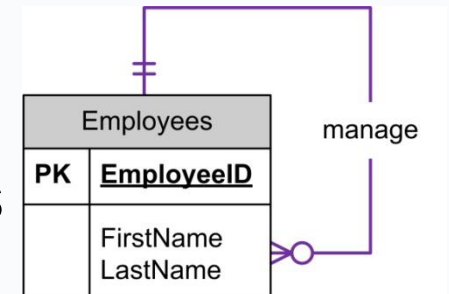
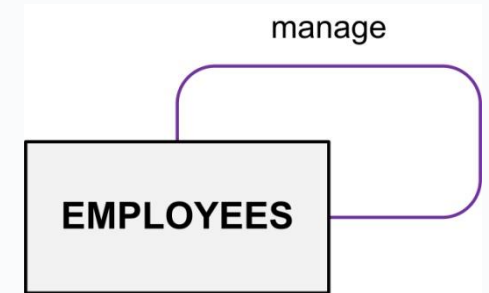
# Degree of a Relationship

- The number of participating entities in a relationship is known as the *degree* of the relationship.



# Unary Relationships

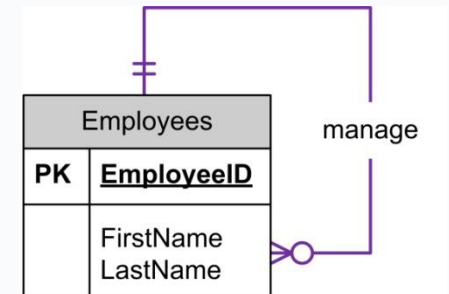
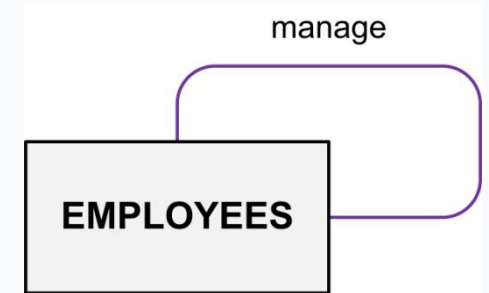
- **Unary** relationships are also known as a **recursive** relationship.
- Instances of entities may have relationships with other instances of the same entity.
- These relationships may be drawn with relationship lines that begin and end connected to the same entity.





# Unary Relationships

- **Unary** relationships are also known as a **recursive** relationship.
  - It is a relationship where the same entity participates more than once in different roles.
  - In the example, employees are managed by employees.
- If you wanted more information about who manages whom, you could introduce a second entity type called manager.

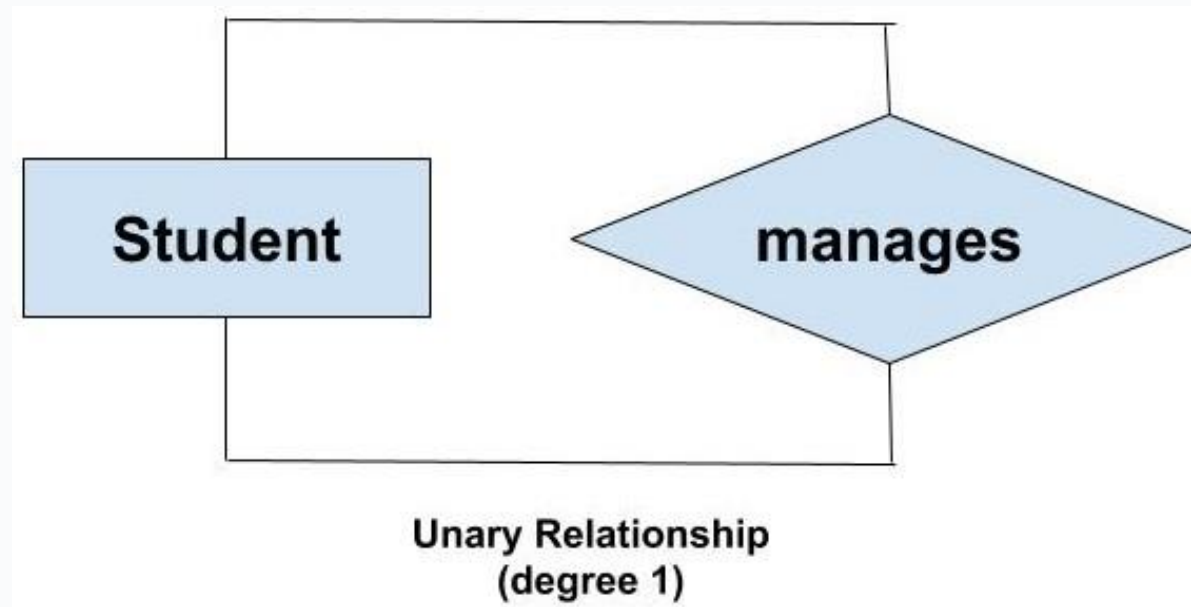


# Example

- ***For example***, Suppose in a classroom, we have many students who belong to a particular club-like dance club, basketball club etc. and some of them are club leads. So, a particular group of student is managed by their respective club lead. Here, the group is formed from students and also, the club leads are chosen from students.
- So, the 'Student' is the only entity participating here.

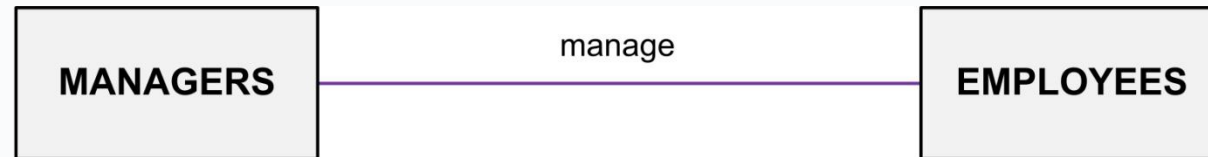
# Example

- We can represent this relationship using the E-R diagram as follows:



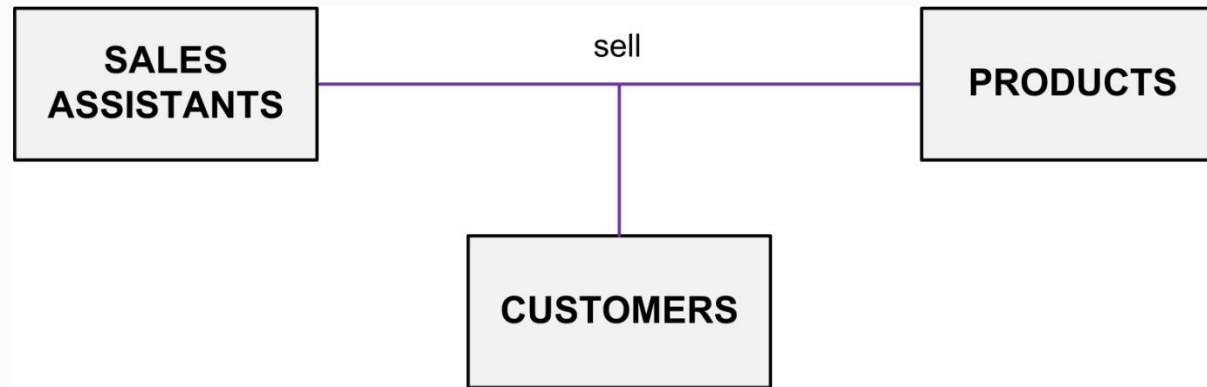
# Binary Relationships

- If there are two entity types involved it is a ***binary*** relationship type.



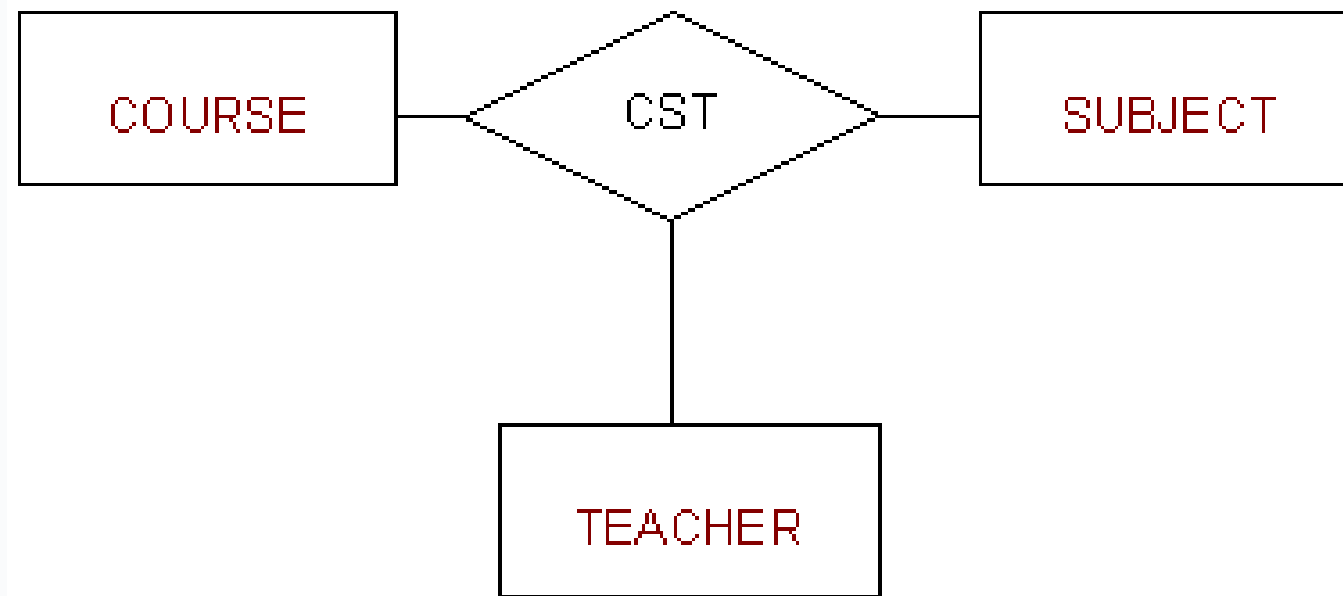
# Ternary Relationship

- If there are three entity types involved it is a ***ternary*** relationship type.

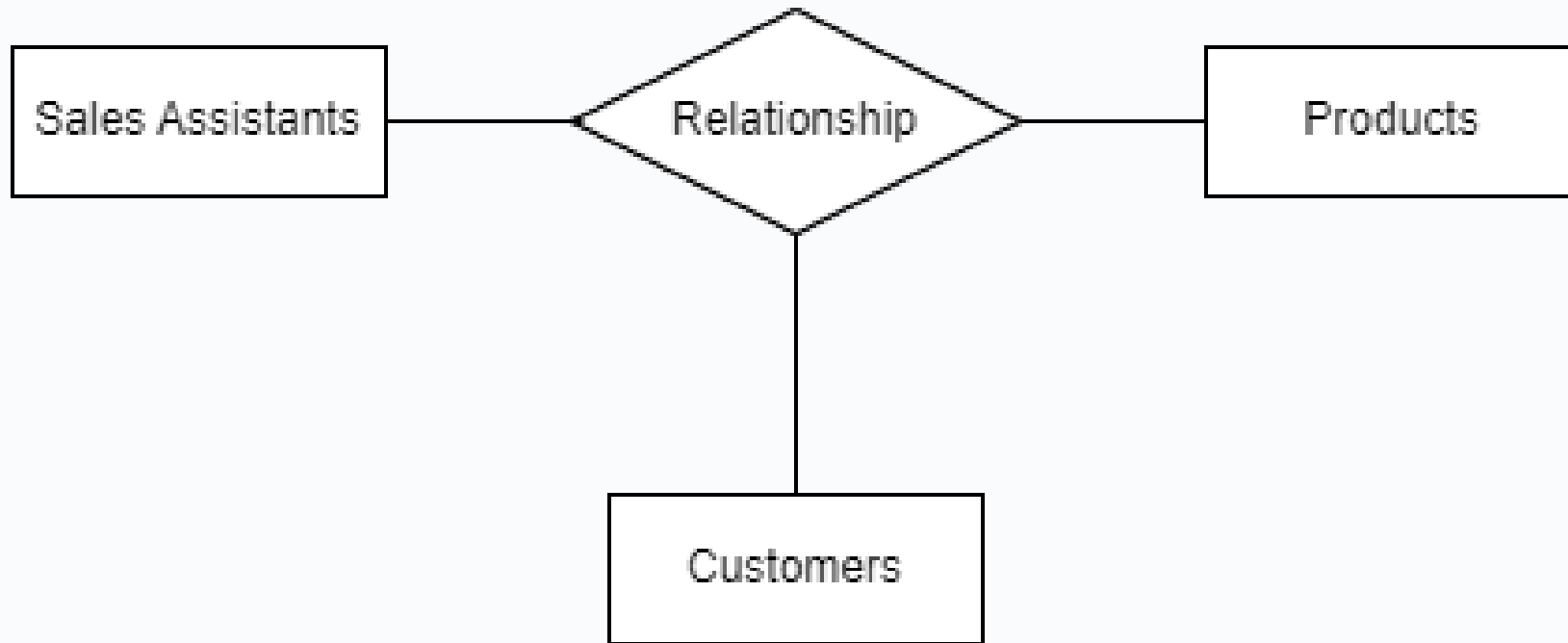


# Example 1

- The University might need to record which teachers taught which subjects in which courses.

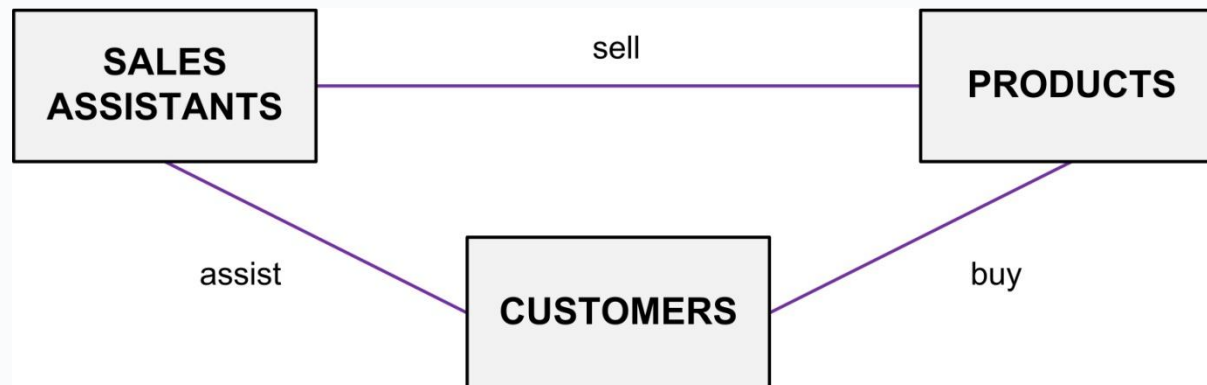


## Example 2



# Replacing Ternary Relationships

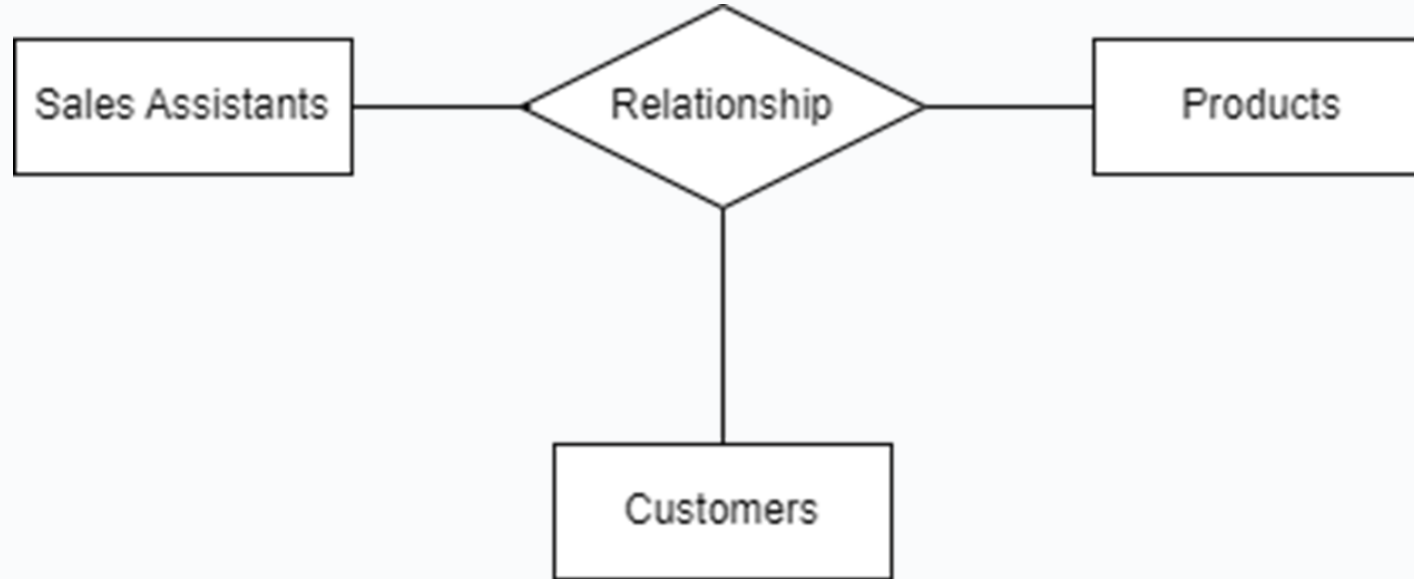
- When ***ternary*** relationships occurs in an ER model they should always be removed before finishing the model.
- Sometimes the relationships can be replaced by a series of binary relationships that link pairs of the original ternary relationship.



This can result in the loss of some information  
i.e. It is no longer clear which sales assistant sold a customer a particular product.



# Replacing Ternary Relationships

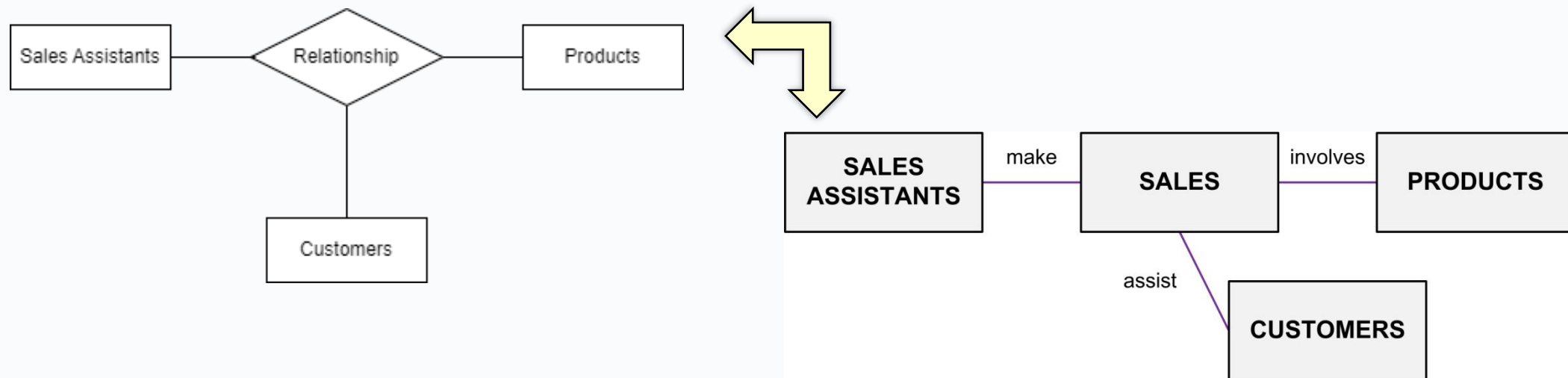


Try replacing the ternary relationship with an entity type and a set of binary relationships.

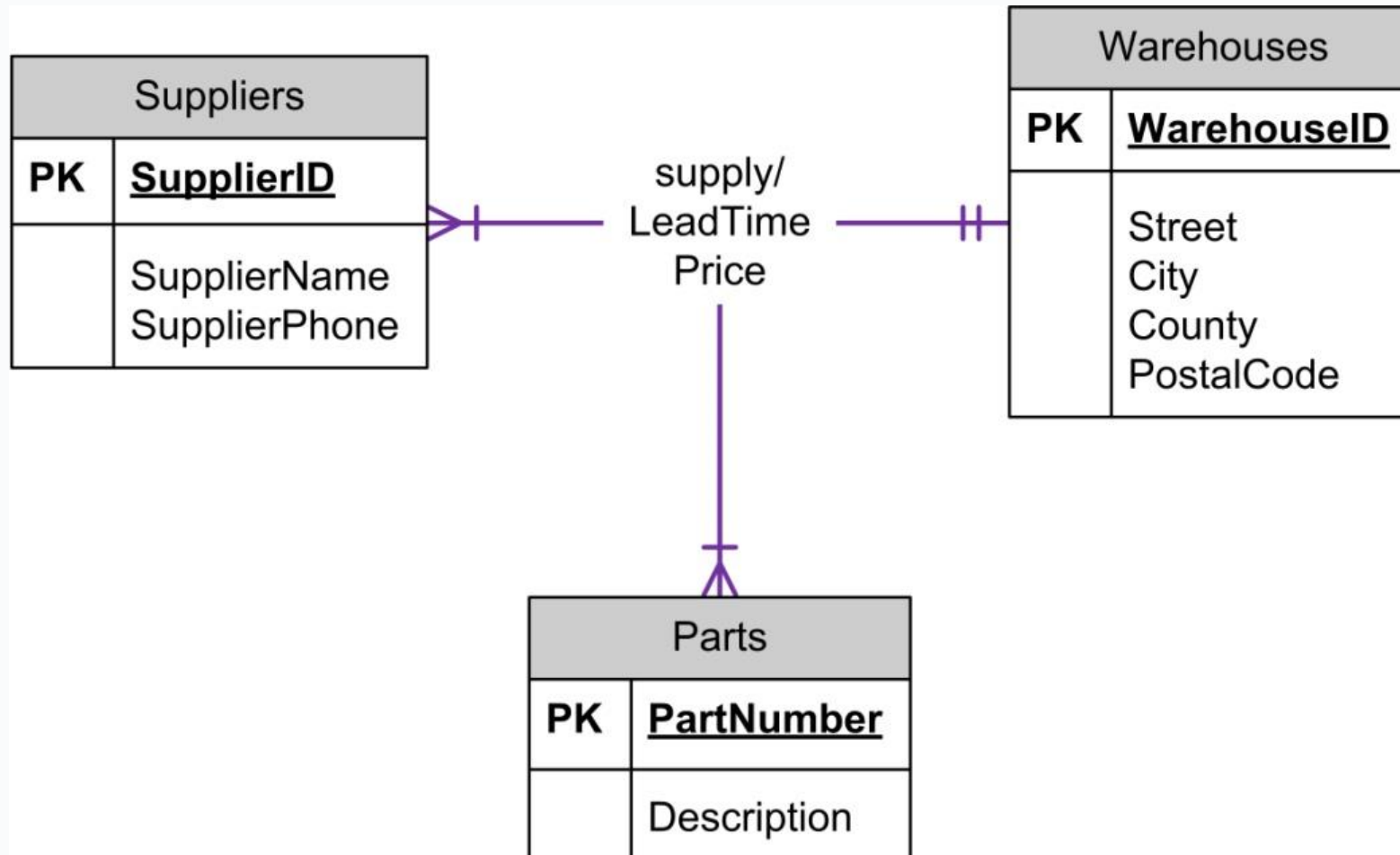
# Replacing Ternary Relationships

Relationships are usually **verbs**, so name the new entity type by the relationship verb rewritten as a **noun**.

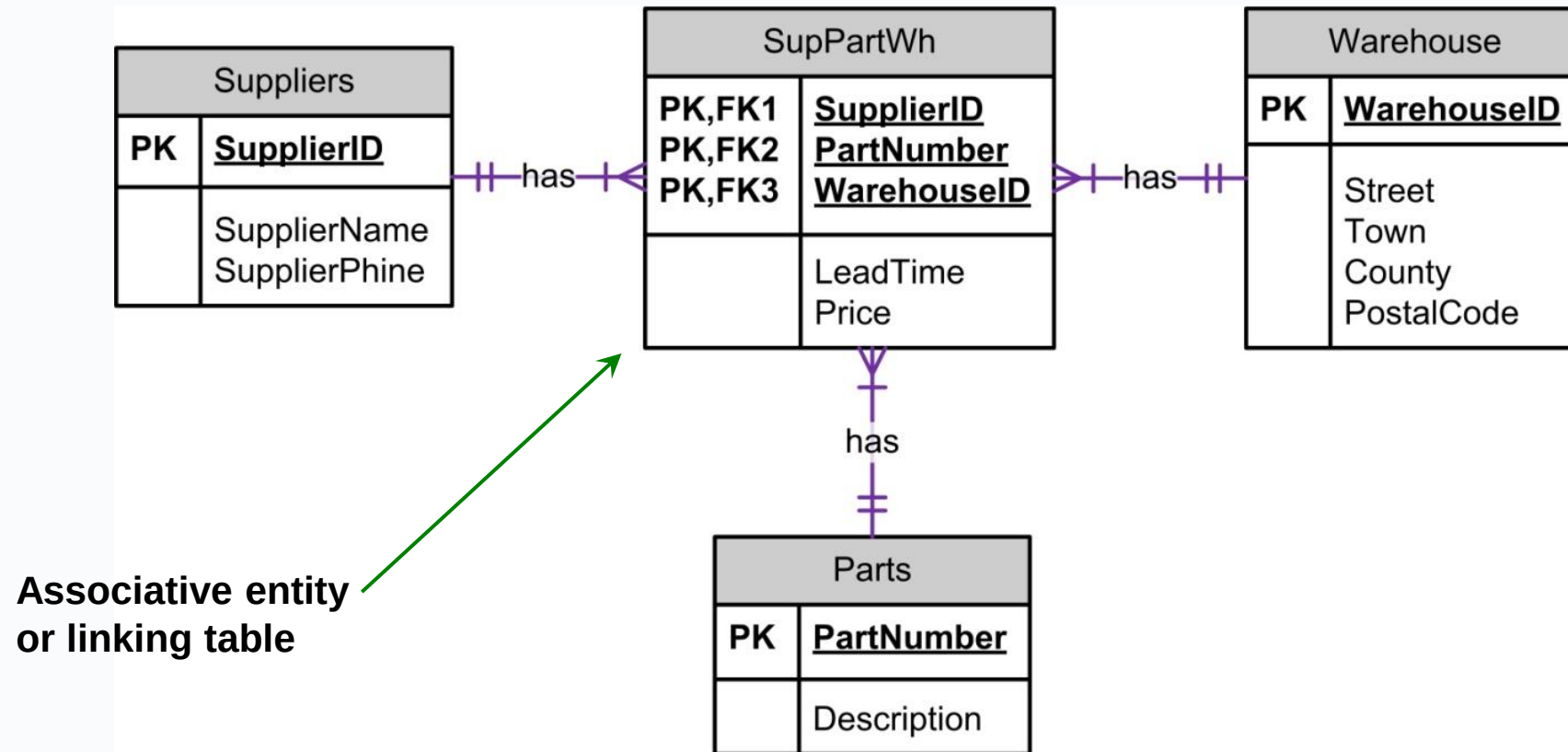
- The relationship 'sell' can become the entity type 'SALES'.



# ERD Example: Ternary Relationships



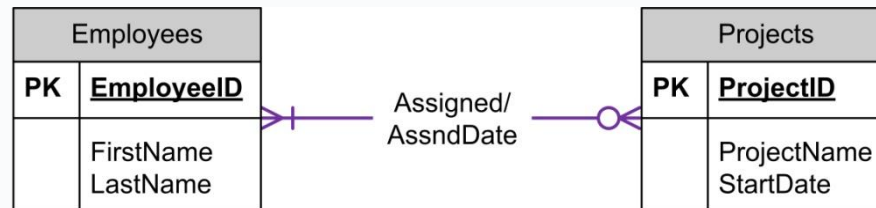
# ERD Example: Ternary Relationship Converted



# ERD Example: Relationship Attributes

## Relationship Attributes:

- Some relationships can have attributes just as an entity can have attributes.
- Generally, this occurs with a many-to-many relationship.



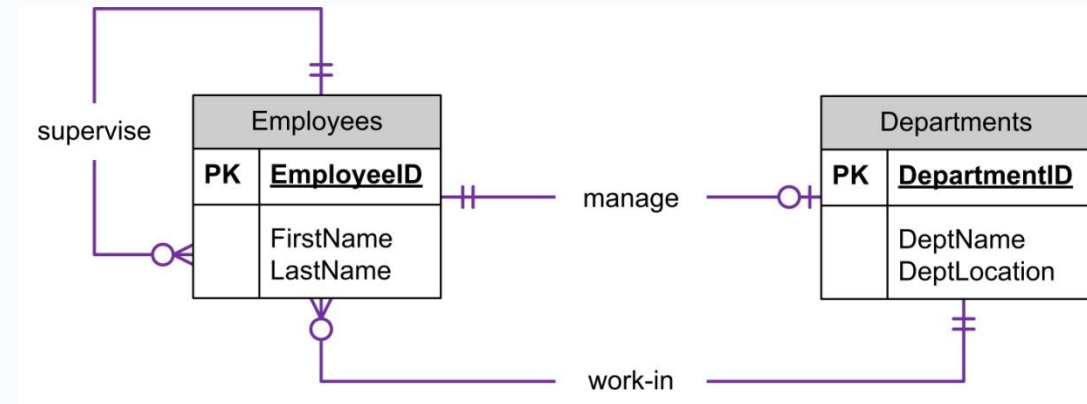
# ERD Example: Relationship Attributes

## Relationship Attributes:

- Rule-of-thumb:
  - If you have an attribute on a 1:N relationship, you probably have something wrong.
  - Attribute should go on one of the entities.

# ERD Example: Multiple Relationships

- **Multiple Relationships** - more than one relationship between the same entity types.
  - These are depicted in the ERD with multiple relationship lines connecting the two entities.



# Solution lab 2 exercise 2

Prepare an ER diagram for a real estate company that lists property for sale. The following describes the organisation:

- The company has a number of sales offices in several counties.
- Attributes of sales office include Office\_Number (identifier/key) and Location.
- Each sales office is assigned one or more employees.
- Attributes of employee include Employee\_ID (identifier/key) and Employee\_Name.
- An employee must be assigned to only one sales office.
- The firm lists property for sale.



# Solution lab 2 exercise 2

Prepare an ER diagram for a real estate company that lists property for sale. The following describes the organisation:

- The **company** has a number of **sales offices** in **several counties**.
- Attributes of **sales office** include **Office\_Number** (identifier/key) and **Location**.
- Each **sales office** is assigned one or more **employees**.
- Attributes of **employee** include **Employee\_ID** (identifier/key) and **Employee\_Name**.
- An **employee** must be assigned to only one **sales office**.
- The **firm** lists **property** for sale.

# Solution lab 2 exercise 2

- Attributes of property include Property\_ID (identifier) and Location.
- Components of Location include Address, City, County, and Postal\_Code.
- Each unit of property must be listed with one (and only one) of the sales offices.
- A sales office may have any number of properties listed.
- Each unit of property has one or more owners.
- Attributes of owners are Owner\_ID (identifier) and Owner\_Name.
- An owner may own one or more units of property.
- An attribute of the association between property and owner is Percent\_Owned.

# Solution lab 2 exercise 2

- Attributes of **property** include **Property\_ID (identifier)** and **Location**.
- Components of **Location** include **Address, City, County, and Postal\_Code**.
- Each unit of **property** must **be listed with one (and only one)** of the **sales office**.
- A **sales office** may have any number of **properties** listed.
- Each unit of **property** has one or more **owners**.
- Attributes of **owners** are **Owner\_ID (identifier)** and **Owner\_Name**.
- An **owner** may own one or more units of **property**.
- An attribute of the association between property and owner is **Percent\_Owned**.

# Entities

## Possible entities

Sales offices

Employees

Properties

Owners

Companies

Locations



## Entities

Sales offices

Employees

Properties

Owners

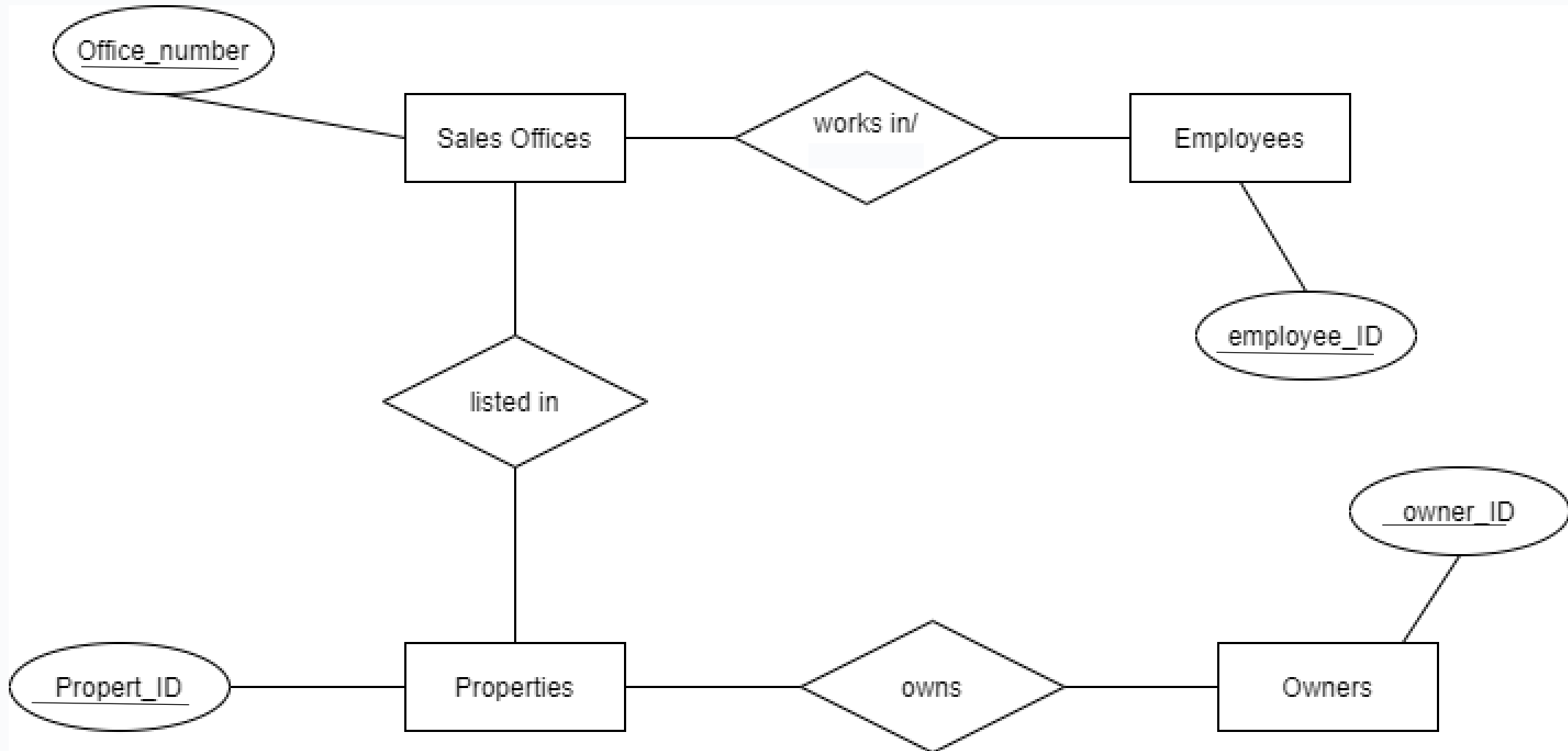
# Attributes

| Entities      | Attributes                                      |
|---------------|-------------------------------------------------|
| Sales offices | Office id, location                             |
| Employees     | Employee id, name, job description              |
| Properties    | Property id, address, city, county, postal code |
| Owners        | Owner id, name                                  |

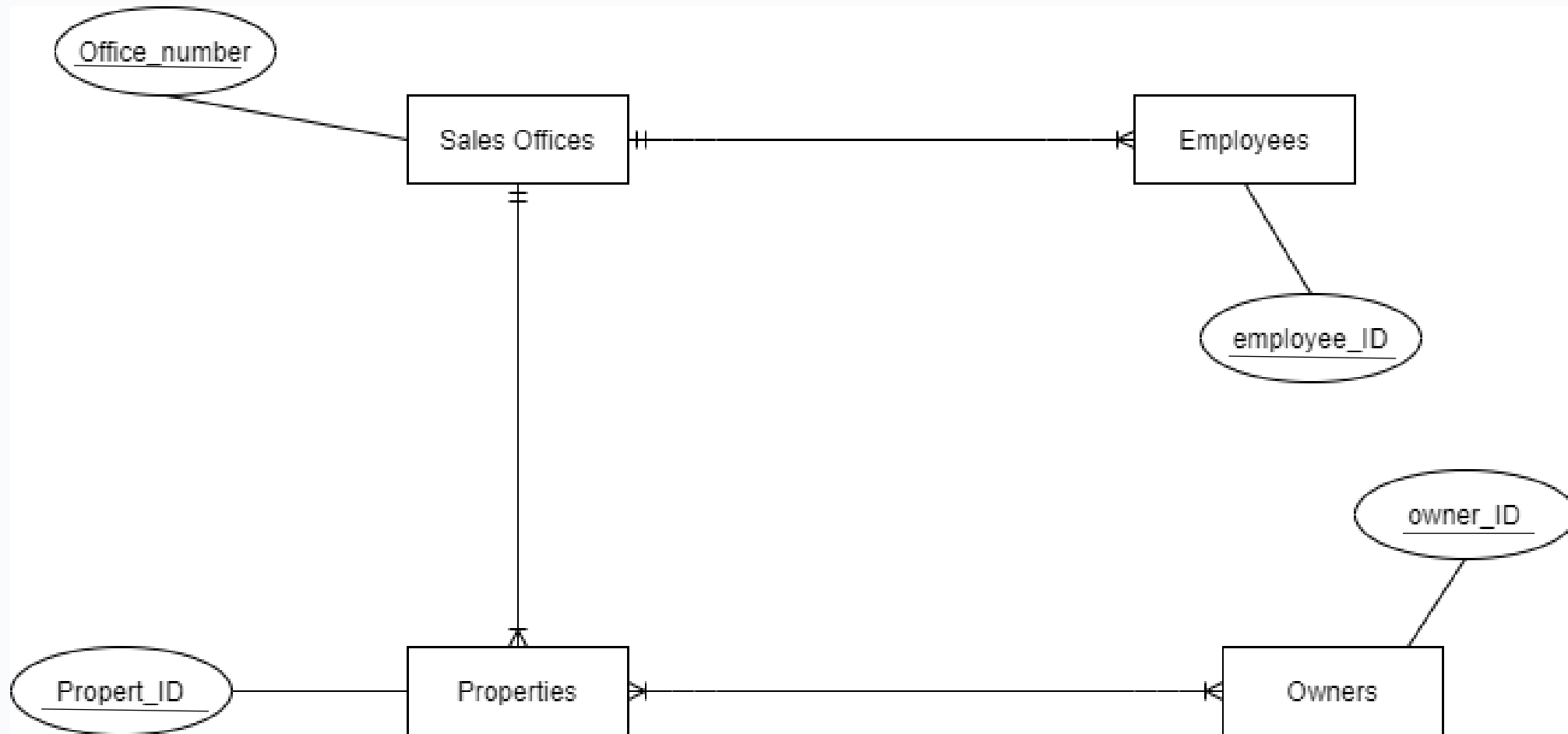
# Relationships

|               | Sales offices | Employees | Properties | Owners |
|---------------|---------------|-----------|------------|--------|
| Sales offices | X             |           |            |        |
| Employees     | Works in      | X         |            |        |
| Properties    | Listed in     |           | X          |        |
| Owners        |               |           | Owns       | X      |

# Rough ER Diagram with PKs

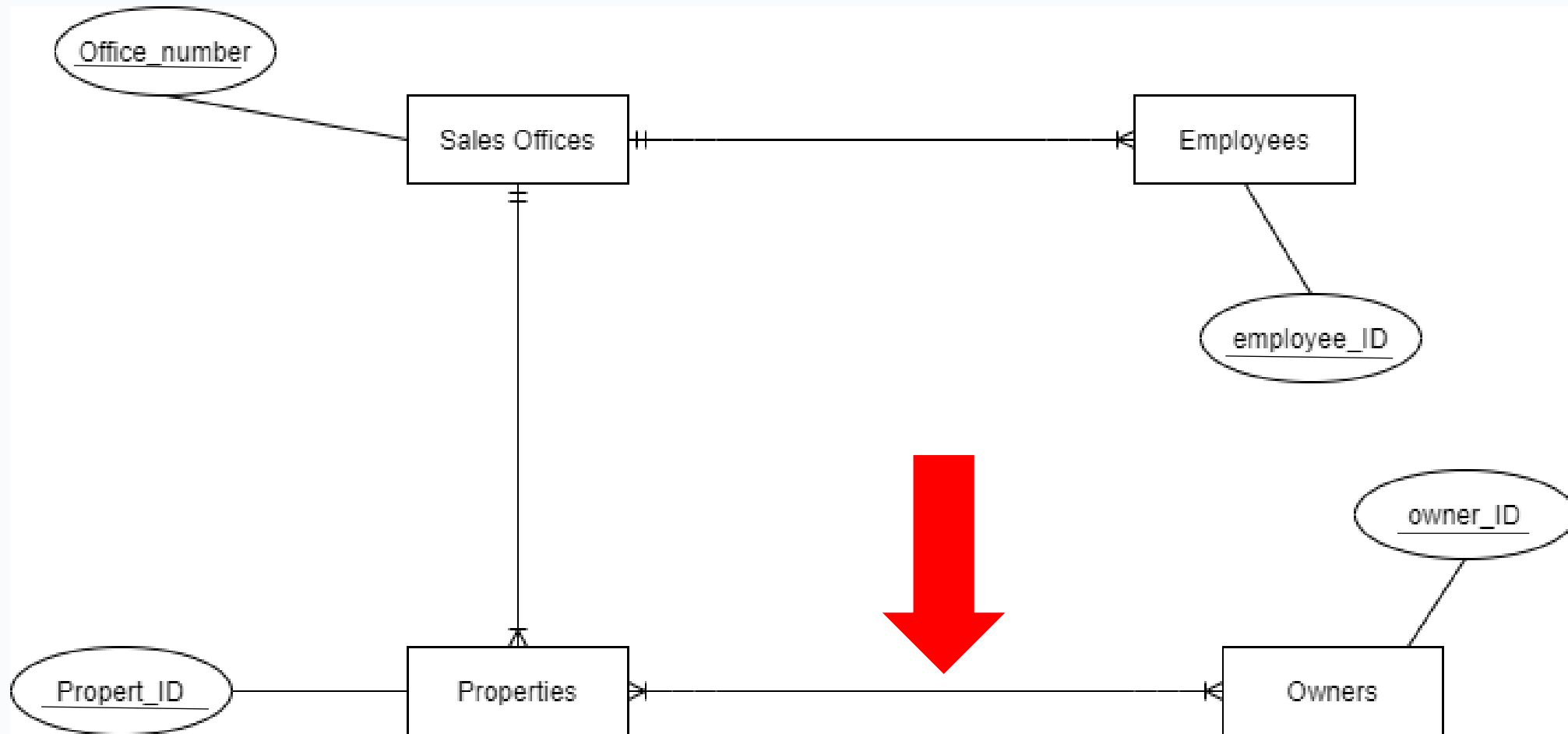


# Rough ER Diagram with cardinality

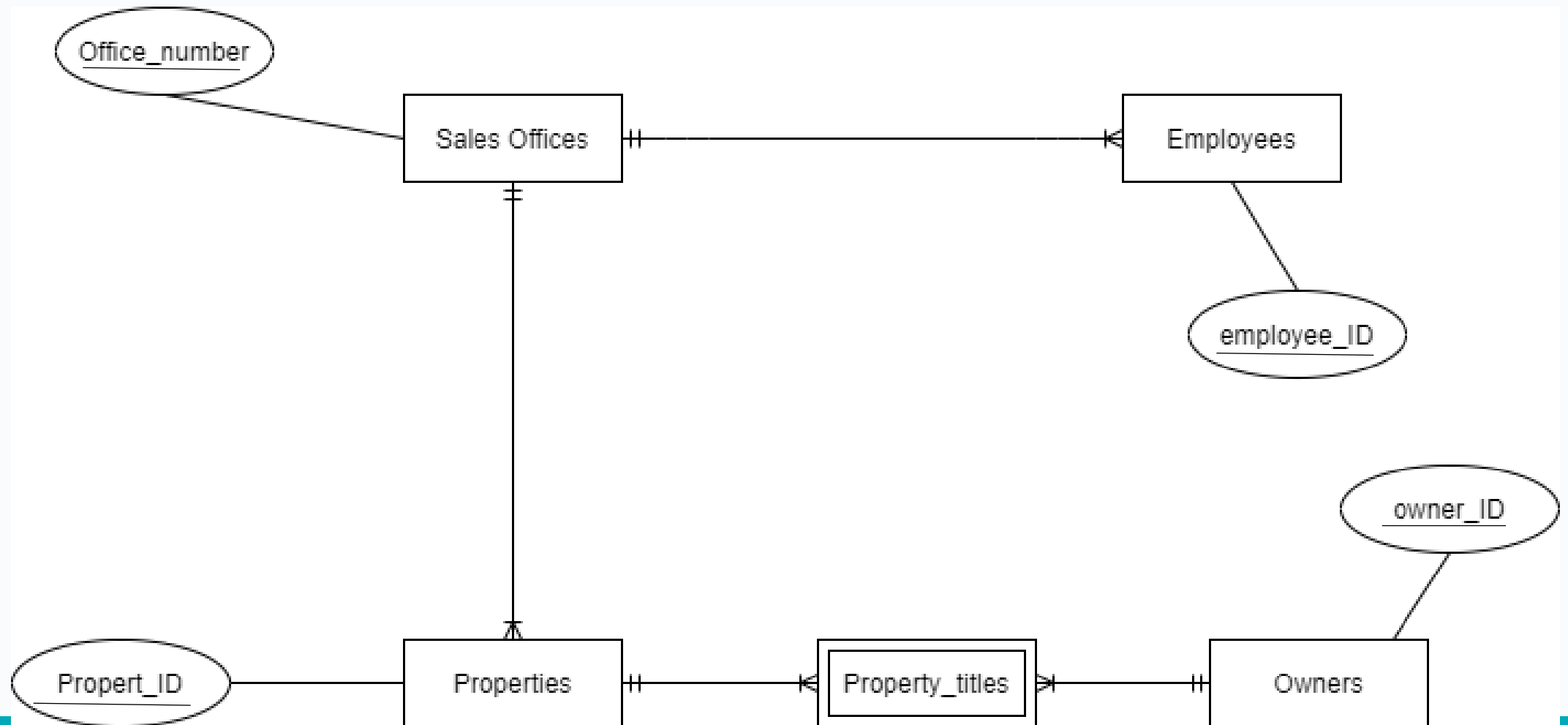




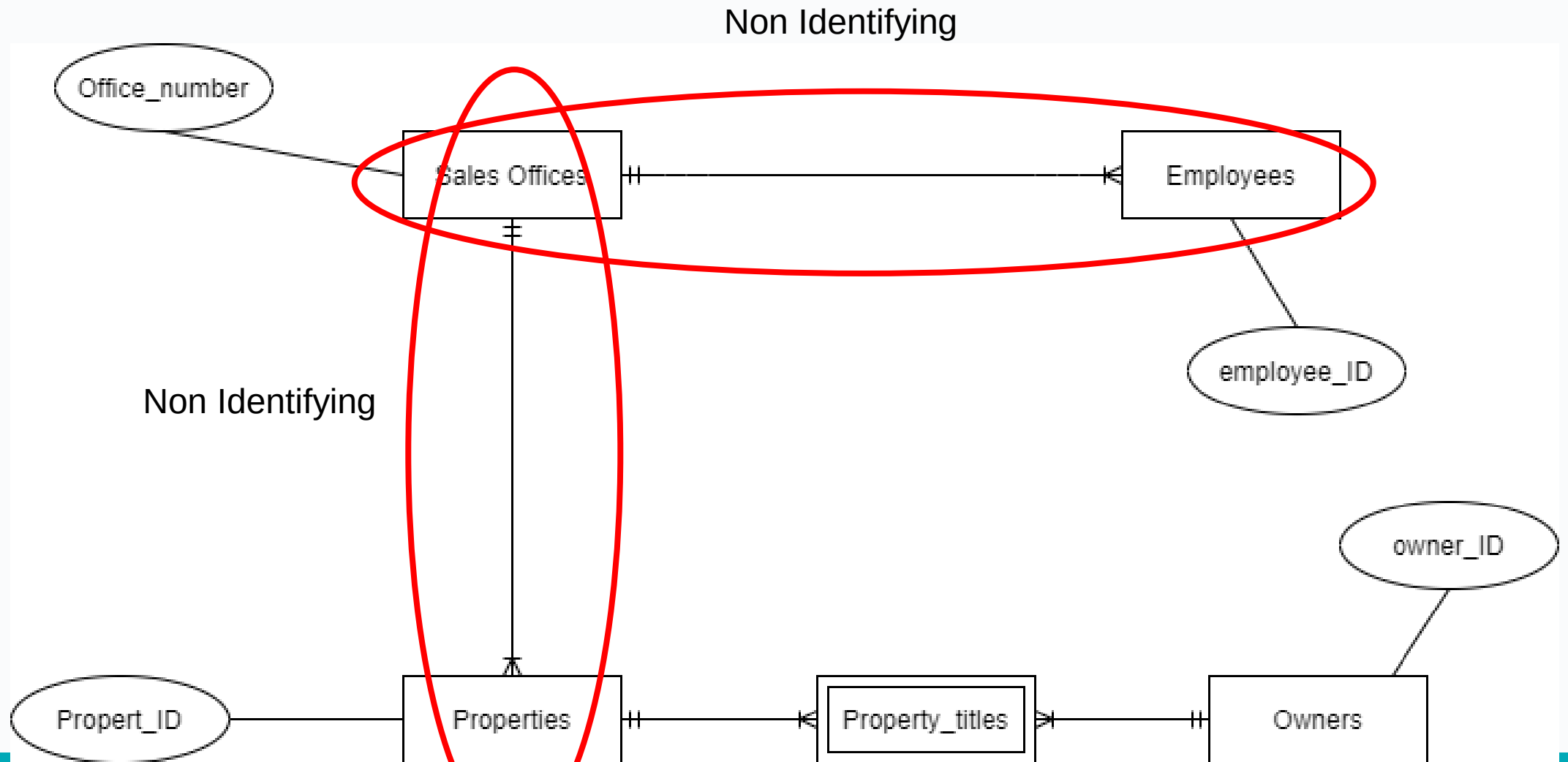
# Rough ER Diagram \*



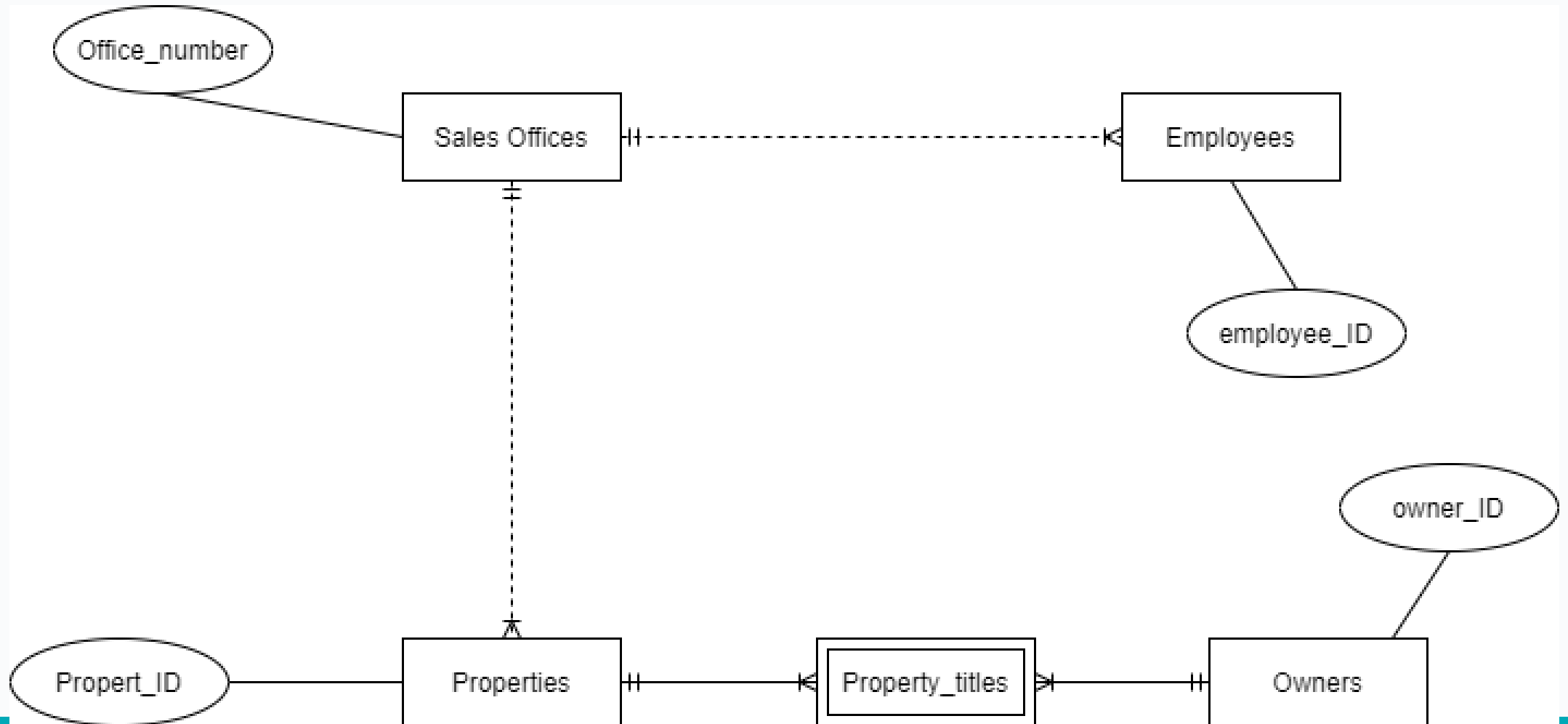
# Rough ER Diagram with weak entities



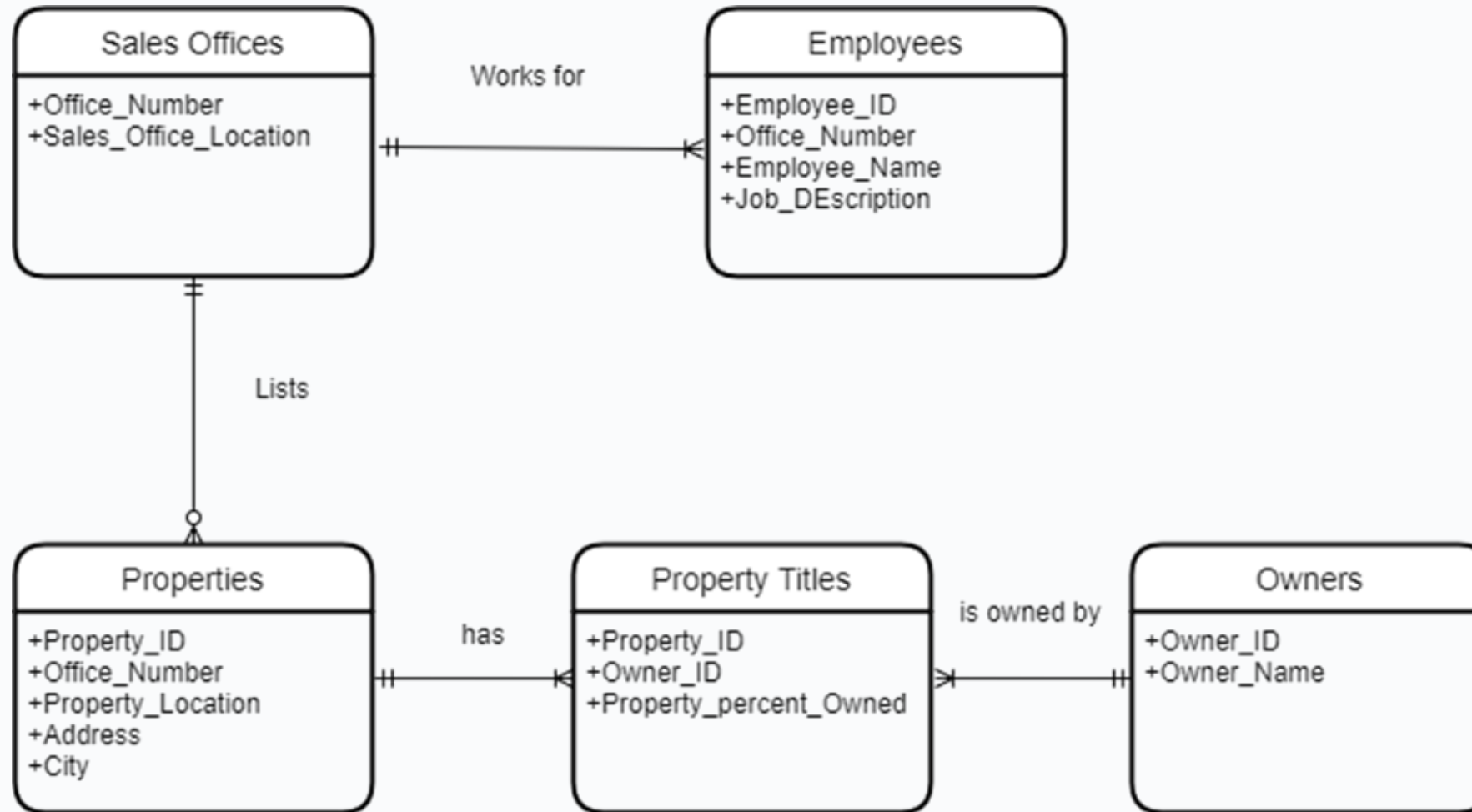
# Rough ER Diagram



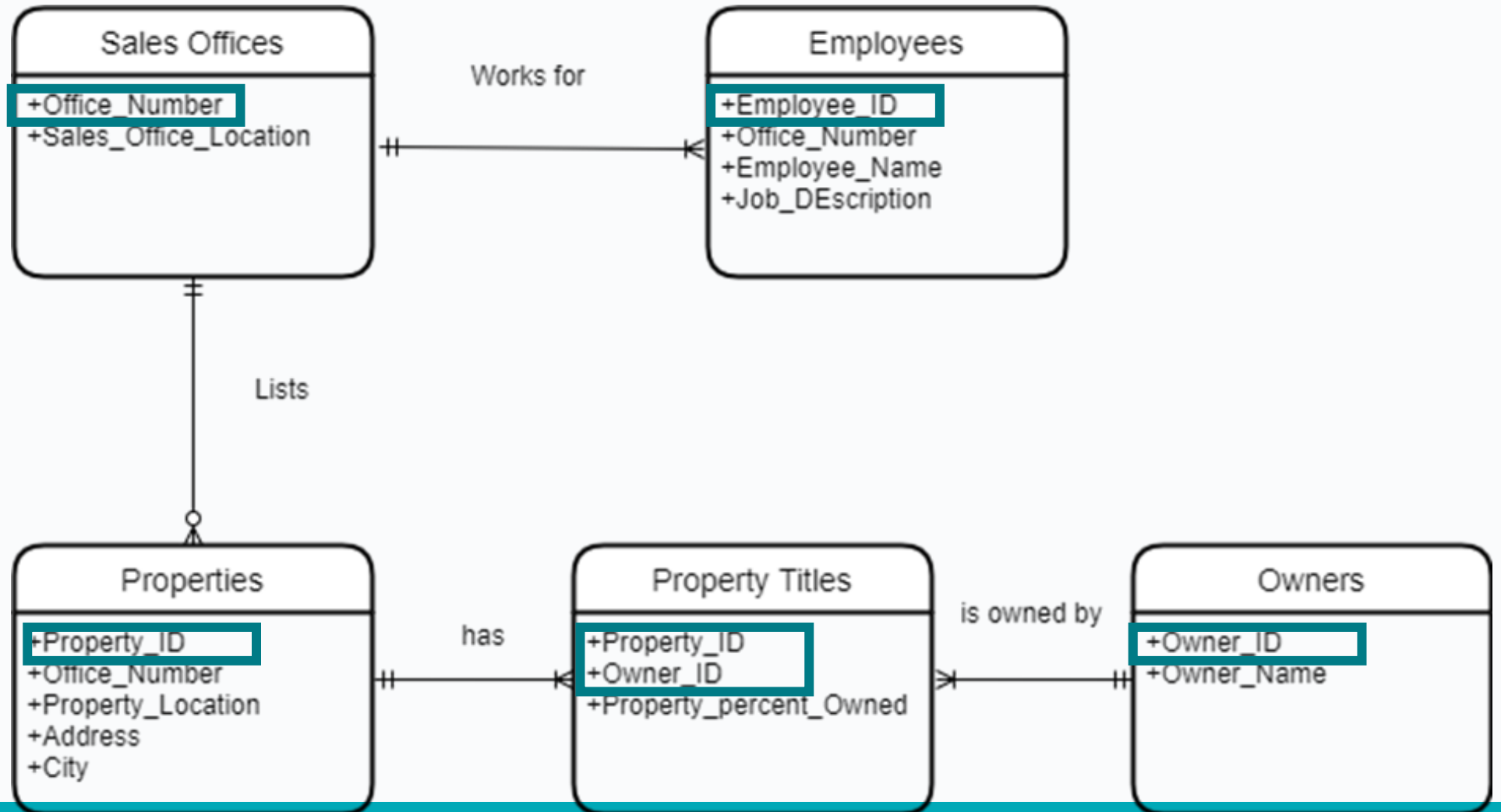
# Rough ER Diagram



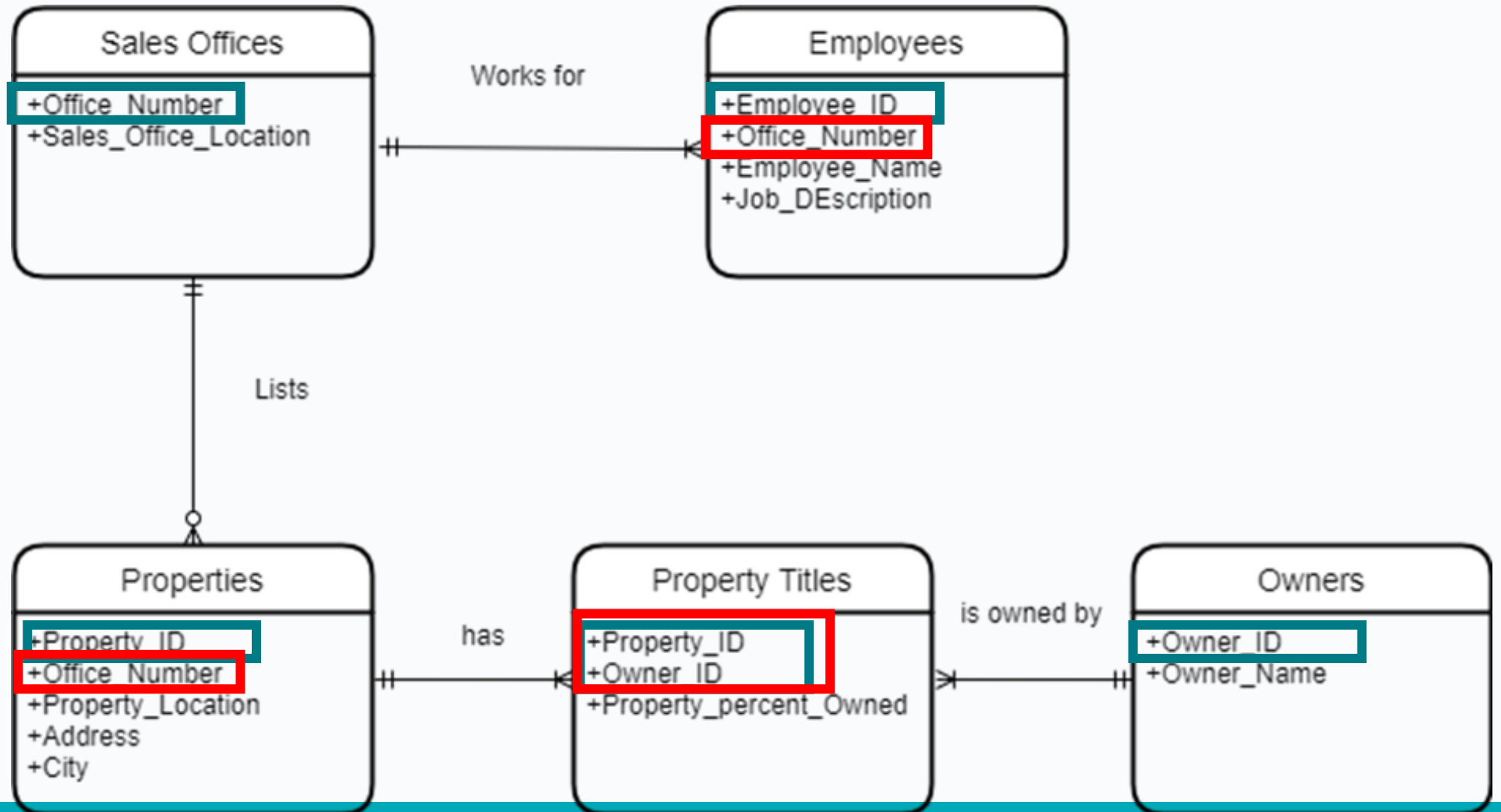
# Solution lab 2 exercise 2 version 1



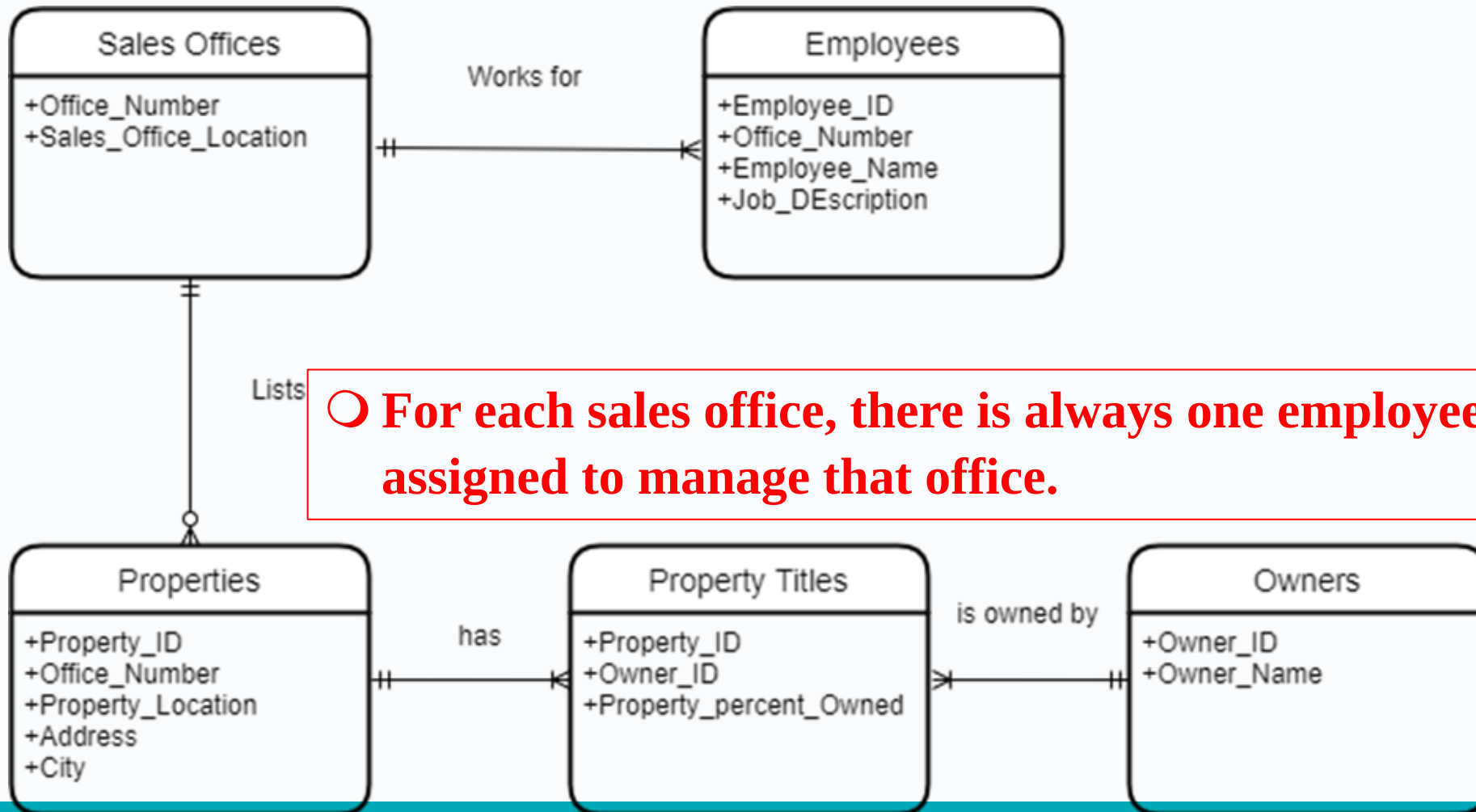
# Solution lab 2 exercise 2 version 1 - PKs



# Solution lab 2 exercise 2 version 1 - FKs

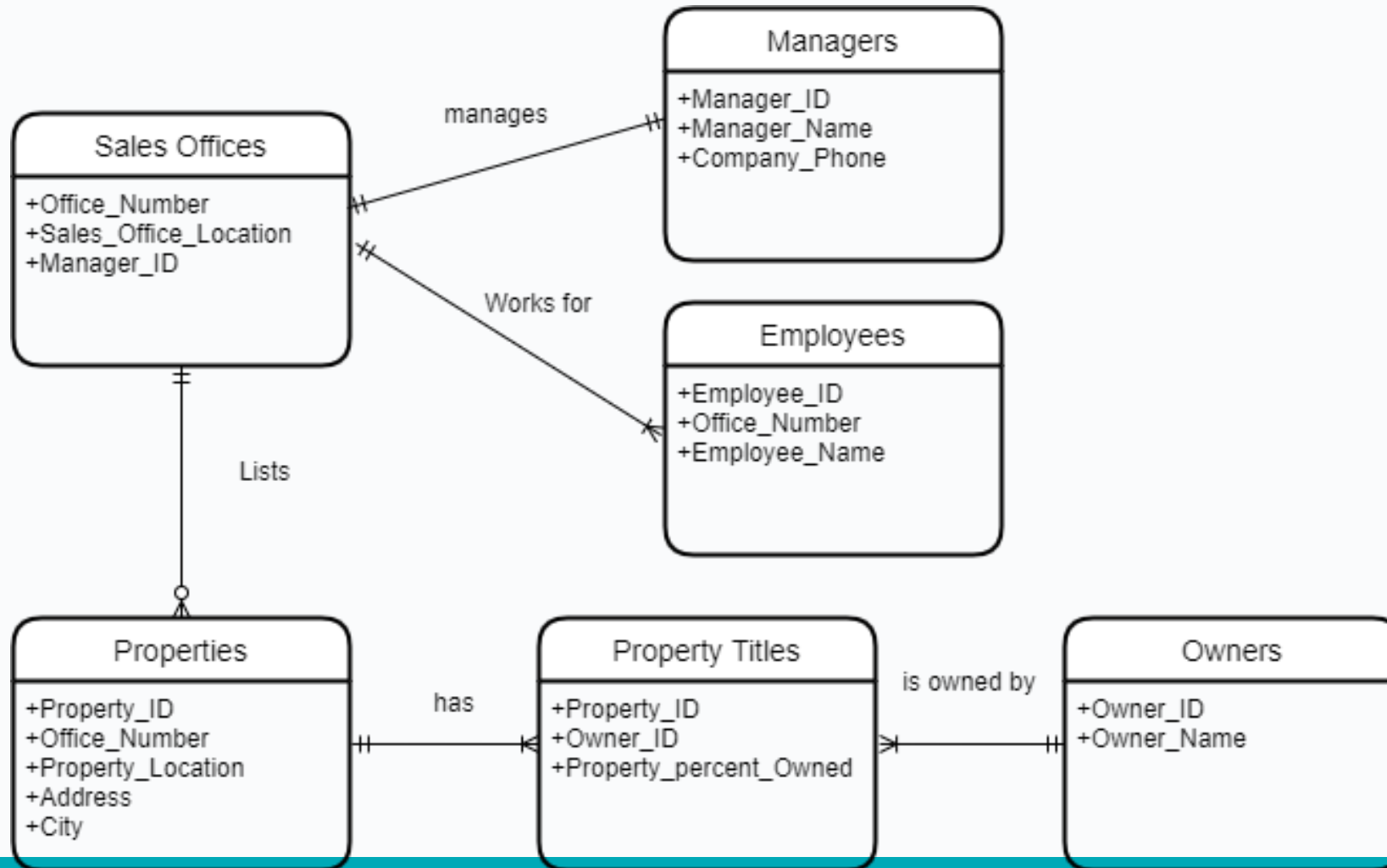


# Solution lab 2 exercise 2 extra information

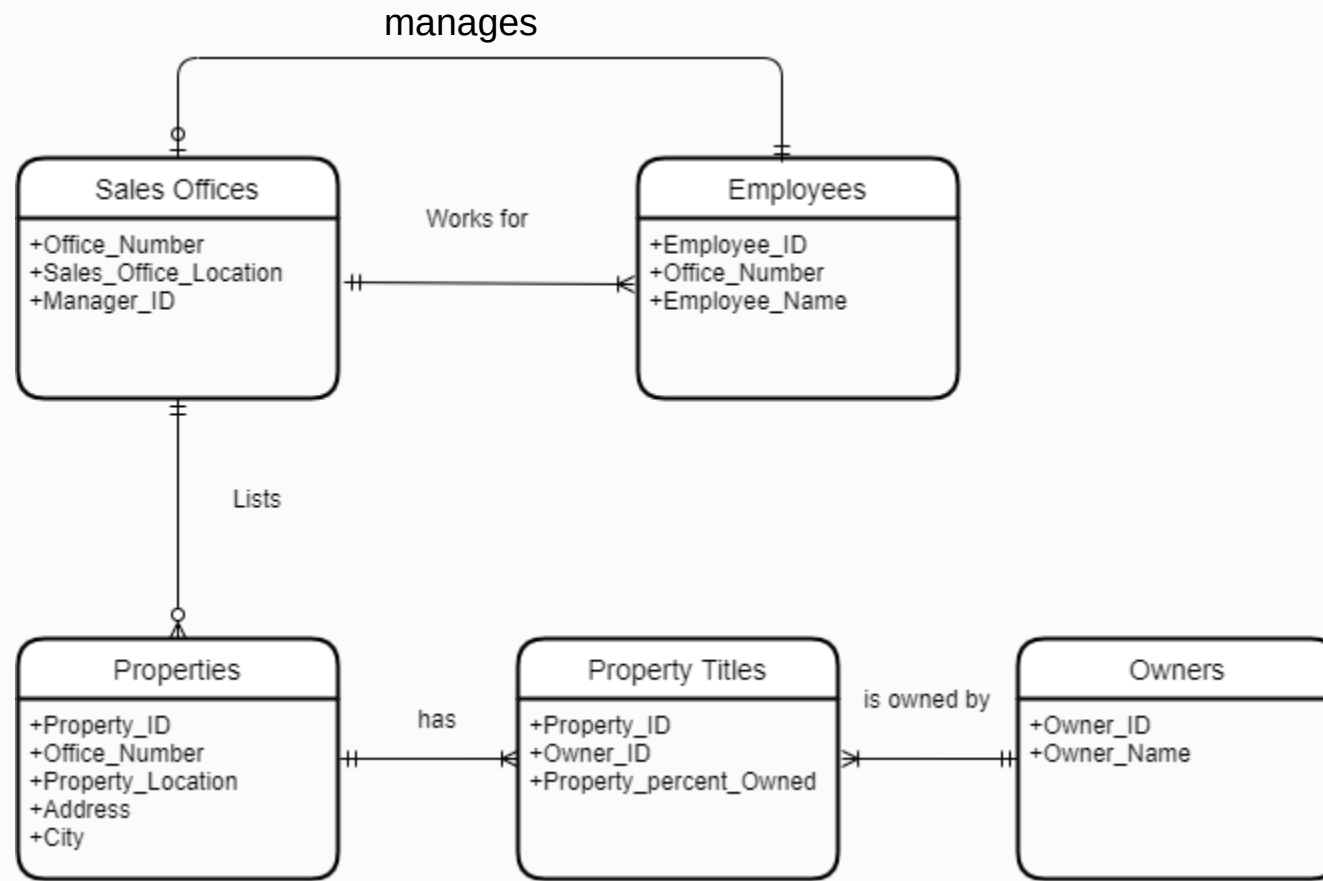




# Solution lab 2 exercise 2 version 2



# Solution lab 2 exercise 2 version 3

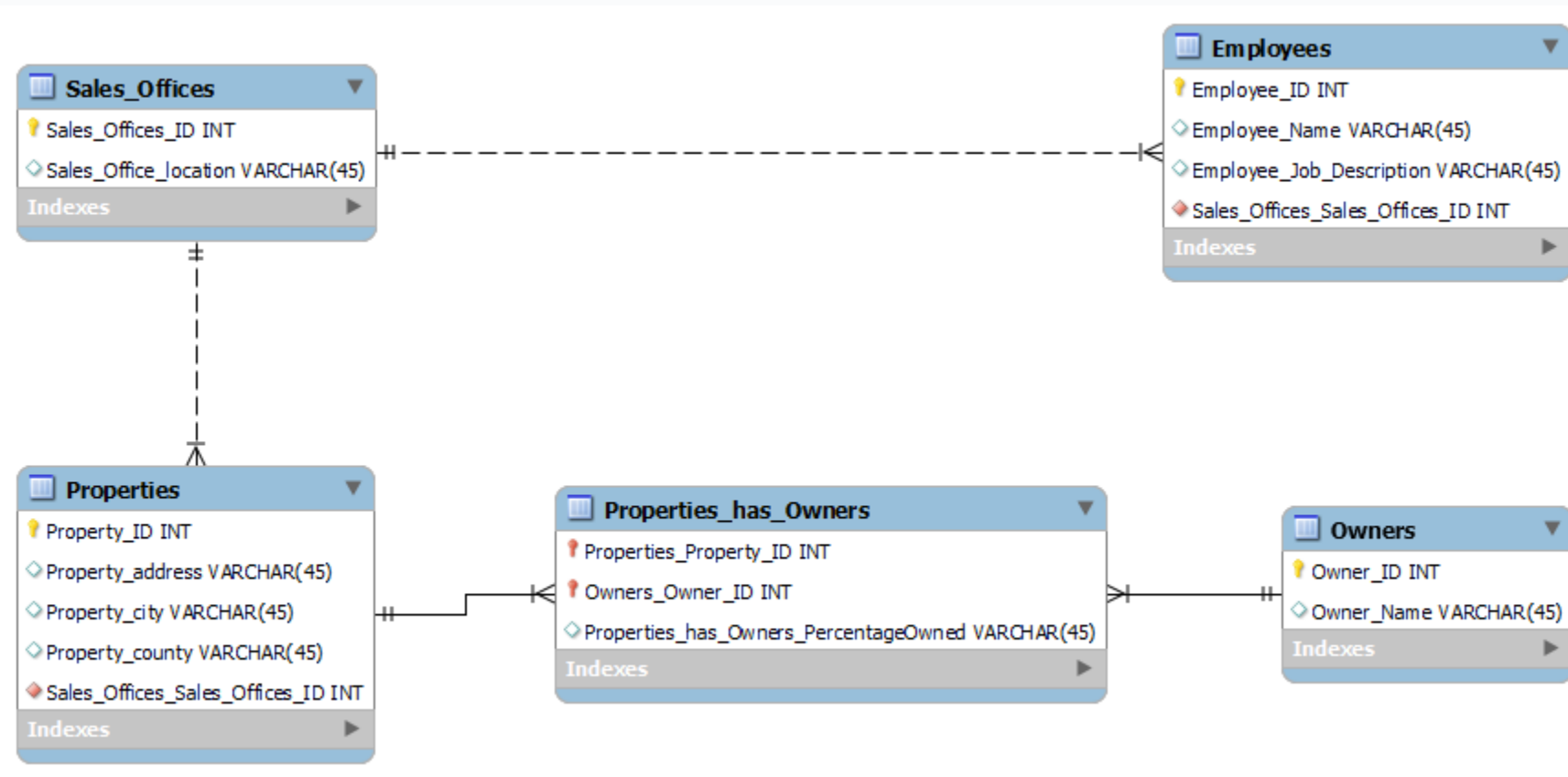


# Key Based ER Diagram

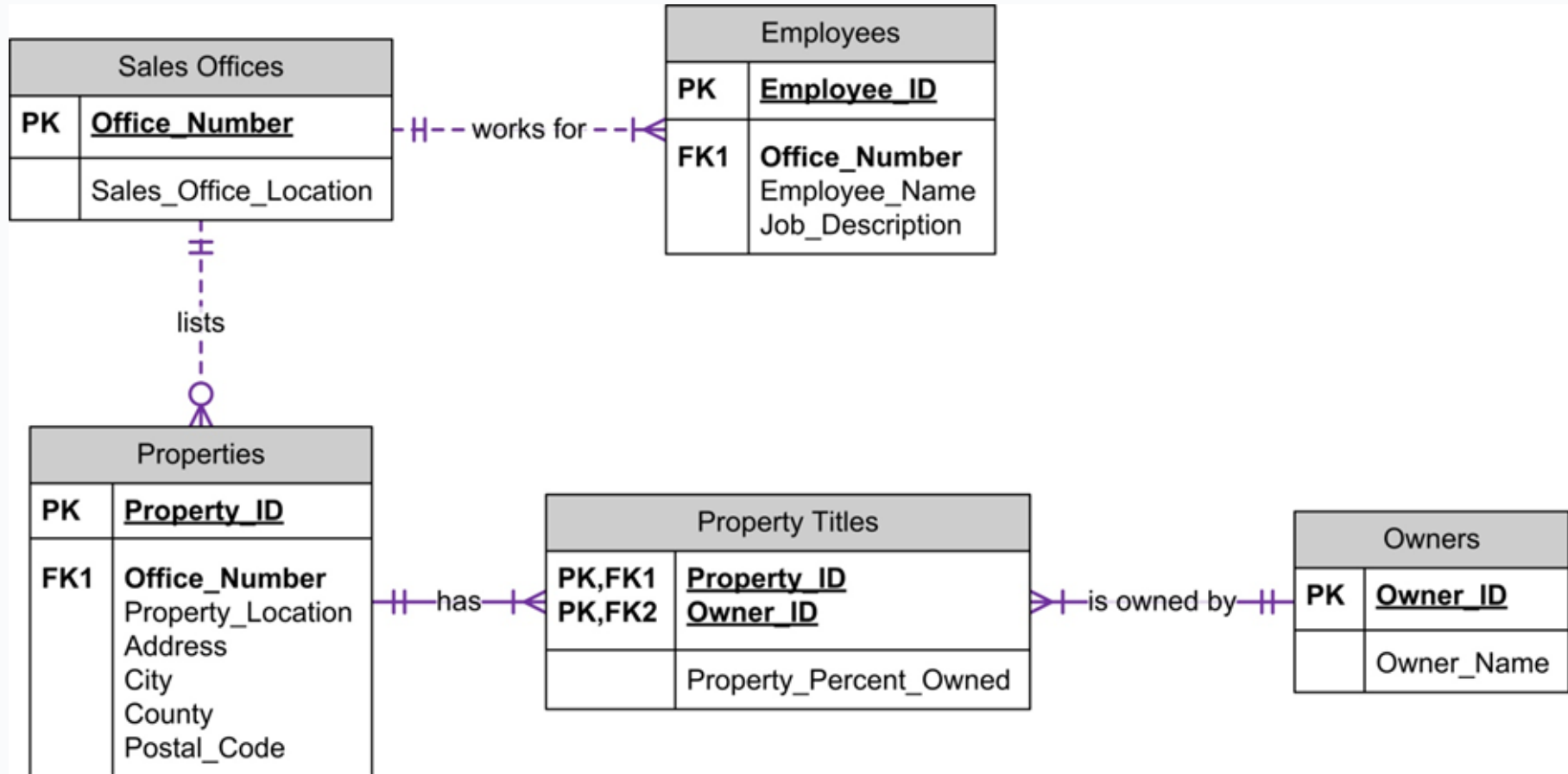
- No many to many relationships
- Bridge entities used if required
- PKs defined
- FKs defined

# Solution lab 2 exercise 2 version 1

## Key based ER diagram

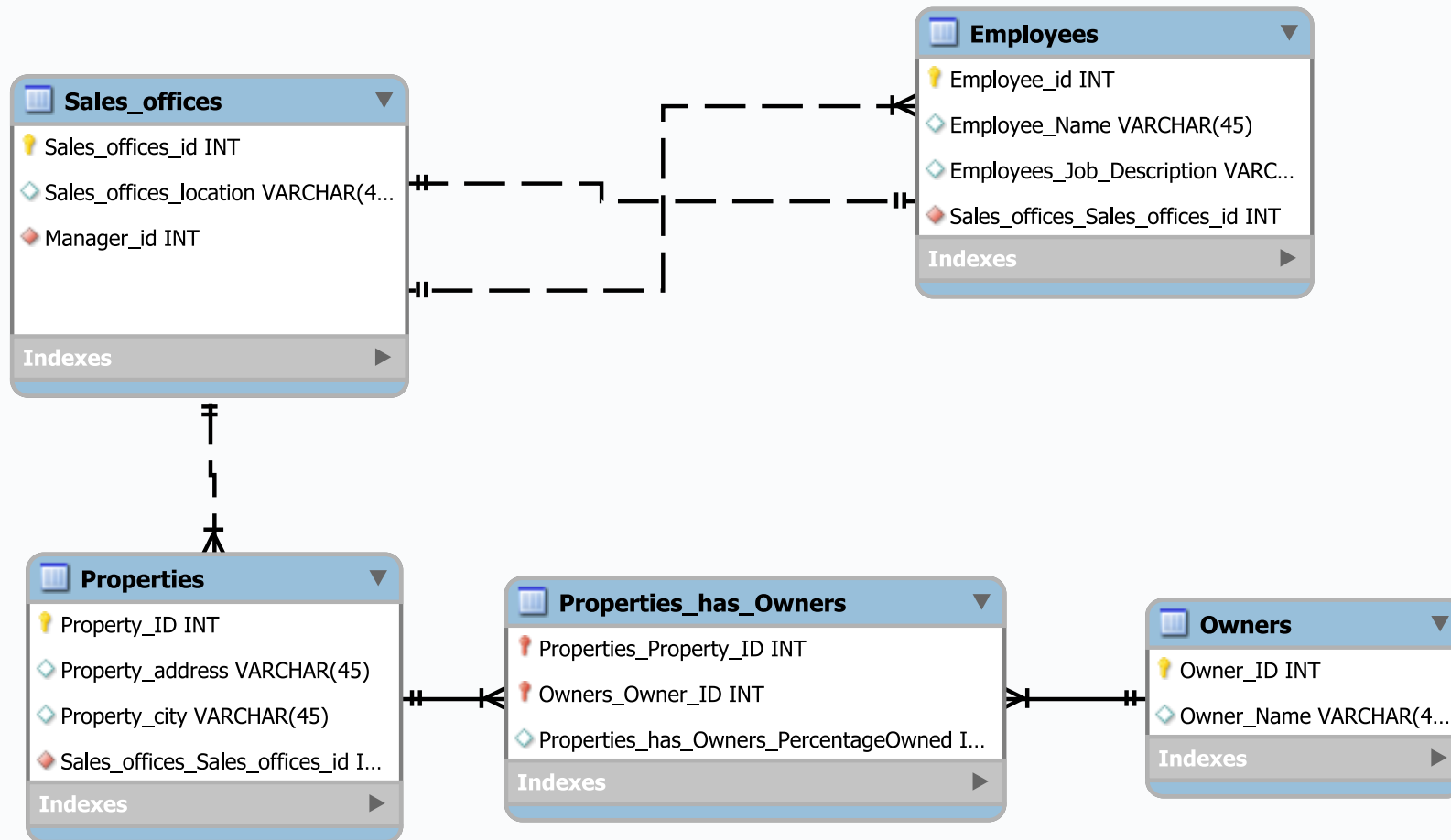


# Foreign Key



# Solution lab 2 exercise 2 version 3

## Key based ER diagram



# Lab 3 description

## Business Requirements

*A company has several departments. Each department has a supervisor and at least one employee. Employees must be assigned to at least one, but possibly more departments. At least one employee is assigned to a project, but an employee may be on holidays and not assigned to any projects.*

# Work flow

Step 1: Identify Entities

Step 2: Find Relationships

Step 3: Draw Rough ER Diagram

Step 4: Fill in Cardinality

Step 5: Define Primary Keys

Step 6: Draw Key-Based ERD (eliminating many to many relationships, include FKs)

Step 7: Identify Attributes & Step 8: Map Attributes

Step 9: Draw Fully Attributed E-R Diagram

Step 10: Revise and update list of assumptions

| Relationship matrix |  |  |  |  |  |
|---------------------|--|--|--|--|--|
|                     |  |  |  |  |  |
|                     |  |  |  |  |  |
|                     |  |  |  |  |  |
|                     |  |  |  |  |  |



# Thanks

- Edmund Nevin for earlier versions of these lecture notes



**Break  
Back at 8pm  
for Lab session**

**TU257  
Information Systems  
Marisa Llorens**